APEX CW

User Manual

Christian Konecny, OE1CKO

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APEX CW – User Manual, for app version 1.8.0 (source 039560f), October 2026. Author: Christian Konecny, OE1CKO. App icon: Sia, OE1LMR.

APEX CW is an independent project. Its training concept and algorithms come from the Morserino-32 firmware by Willi Kraml, OE1WKL, and the Morserino-32 team; there is no other connection to them.

This manual is also available in German.

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1 Introduction

APEX CW is an Android app for learning and practicing Morse code (CW): copy training with the Koch method, sending practice with the Echo Trainer, a CW Keyer, a CW Decoder using the microphone, CW over the internet (WiFi Trx), a QSO Bot and several games. An **adaptive block flow** tracks your mistakes character by character and suggests when you are ready for a new character, more speed or shorter pauses.

Where the app comes from

APEX CW is an independent hobby project by Christian Konecny, OE1CKO. Many ideas and much of the training logic come from the open-source firmware of the Morserino-32 by Willi Kraml, OE1WKL. Its training concept, with the Koch sequences, the Echo Trainer, the QSO Bot and much more, is the result of years of careful work by Willi and his team. Many thanks for that!

The logic was read out of the firmware source and rewritten for Android; the app contains no original firmware code. Beyond that there is **no connection** to Willi Kraml or the Morserino-32 team, and the app is not a product of the Morserino project.

The app was based on firmware version 9.0.0. Later firmware changes are not carried over automatically.

About this manual

This manual describes the app version printed on the title page. You can see which version you have installed under **Settings → Info** (see Info: version and build).

Settings that also exist on the Morserino have plain names in the app (for example **Character spacing** instead of “Interchar Spc”). The table Morserino terms lists which Morserino menu item belongs to which setting.

If you are new, read the chapter Getting started: learning Morse step by step first: a practice plan that puts the app’s features into a sensible order.

2 Getting started

2.1 Installation

The app is currently handed out as an APK file, not through the Play Store. It needs Android 8.0 or newer and a 64-bit device (practically every phone since about 2017); it cannot be installed on 32-bit-only devices.

1. Copy the APK file to your phone, for example by messenger, e-mail or USB cable.
2. Tap the file in your file manager. The first time, Android asks whether the app you install from may **install unknown apps**. That might be your file manager or browser. Allow it for that app.
3. Confirm the installation.

Updates work the same way: install the new APK over the old one, and your settings and statistics are kept. This only works if both APKs are signed with the same key. If Android reports a conflict, you first have to uninstall the old version, and that deletes your settings and statistics.

2.2 The home screen

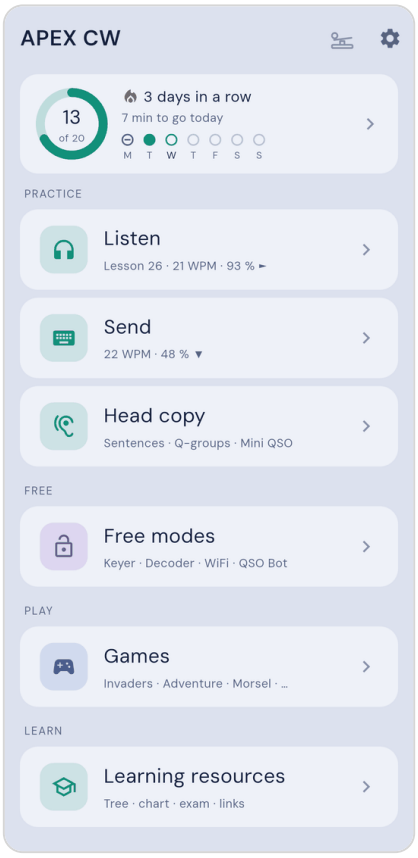
The home screen has four groups:

Group	Tile	What for
Practice	**Listen**	Practice copying: CW Generator and Koch Trainer in the block flow
	Send	Practice sending: Echo Trainer. A word is played and you key it back
	Head copy	Listening without writing: sentences, Q-groups, Mini QSO (see Head copy)
Free	**Free modes**	Opens five tiles: **CW Keyer** (key freely, with the text decoded on screen), **CW Decoder** (copy CW through the microphone), **WiFi Trx** (CW over the internet with other Morserinos and apps) **QSO Bot** (a simulated QSO partner) and **Own texts** (hear your own texts from the clipboard as Morse, see Own texts)
Play	**Games**	Morse Invaders, text adventure, Morsel, Memory Chain, Trailblazer, Fox Hunt, Fight the Pileup, Radio Cave
Learn	**Learning resources**	Interactive Morse tree, character chart, exam simulation, links to courses and practice sites

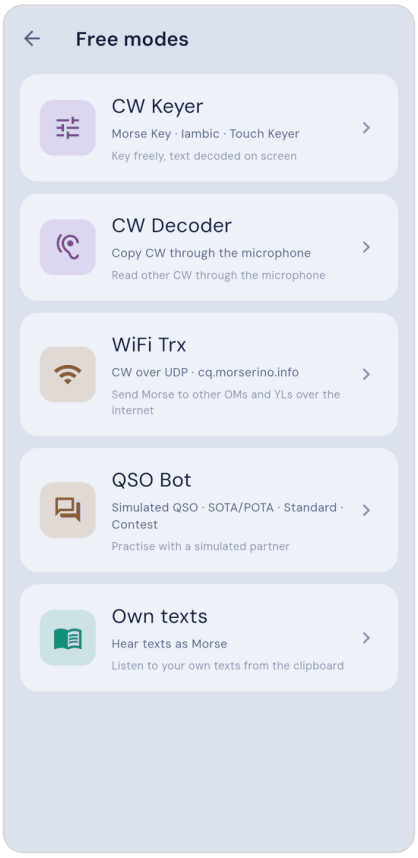
At the very top sits the **daily goal card**: a ring with your active practice minutes today, your streak, and the week Monday to Sunday as dots. Tap it to open the Achievements page.

The **Listen** and **Send** tiles show your current Koch lesson and speed, and – from the sixth block on – the trend of your last blocks (see Trend).

The gear icon at the top right opens the **global settings** (chapter Settings). Anything that concerns only one training is set up directly inside that training.



The home screen



Free modes: five tiles

2.3 Controls inside the trainings

Each training has these icons at the top right:

- ⚙ **Training settings** opens a sheet that slides up from the bottom. Every change takes effect immediately and is saved. Shortcut: a **long press** on the **Koch** or **Words** chip selects it and opens the sheet right at its settings (Koch sequence, or word selection). A short vibration confirms the long press, and a back arrow at the top of the sheet takes you straight back.
- **Statistics** (only in **Listen** and **Send**) shows your progress per character (see Character statistics).

While a block is running, these icons and the selection at the top are hidden to keep the screen calm. The **back button** then only ends the running block and returns you to the training's start view. A second back leaves the training.

The screen stays on while you practice.

Text size: you can **pinch** text areas that show decoded text or results to make the text larger or smaller. This works in the CW Keyer, in Send and on the result page, among others. Each area remembers its own size.

2.4 Light, dark, language

Under **Settings → Appearance** you choose the **Theme** (System, Light, Dark, Amber, Red, Green, Warm grey or Navy & orange) and the app **Language** (Deutsch or English; without your own choice it follows the phone's language). Amber, Red, Green, Warm grey and Navy & orange are dark themes with their own palette: Amber with warm yellows, Red (night vision) with muted red, which disturbs dark adaptation the least, Green like a phosphor terminal, Warm grey with a soft, low-contrast brown-grey, and Navy & orange with a blue-black ground, silver text and an orange accent (only orange and silver tones, like a smartwatch face; all menu icons are orange). There is also a **Blue light filter** you can switch on: it tints the whole app warm (in light and dark) at a strength you set. All of it takes effect immediately.

The app follows the Android **font size** setting, but only up to **1.3×** the default size (step 4 of 7 on a Pixel). Larger steps look the same as that one in the app; otherwise the training screens would no longer fit. The on-screen keyboard for typing along and the Koch character row always keep their size. The Android **Display size** setting ("Make everything bigger or smaller") has no effect in the app; it always uses the device's default size. When a page is longer than the screen, a scrollbar at the edge shows that there is more.

2.5 Sound and volume

Set the volume with the phone's volume keys. You set the **pitch** and the **tone softness** in the settings. When headphones, a USB audio device or a Bluetooth device is connected or disconnected, the app switches to it automatically. If you don't want that, you can fix the output (see Audio output). How to practice with noise, fading and a neighbouring station is described under Interference.

Bluetooth headphones usually add noticeable latency. That doesn't matter for listening, but it does for sending: you hear your sidetone noticeably later than you key. For sending, use wired headphones or the speaker. Wherever the keys (dit/dah or straight key) are shown, a small line above them warns you while the sound plays over Bluetooth. Tap the line to open the settings right at the audio output, so you can switch there. Close it with the X until the next app start; turn it off for good in the settings under Audio output.

3 Basics

3.1 Speed in WPM

Speed is given in **WPM** (words per minute), based on the standard word “PARIS”. At *w* WPM a dit lasts exactly 1200 / *w* milliseconds, so 60 ms at 20 WPM.

3.2 Spacing in dits: character spacing and word spacing

As on the Morserino, the pauses are set **in dit lengths**:

- **Character spacing** is the pause between two characters of a word. Normal Morse is 3 dits. You can set 3 to 45.
- **Word spacing** is the pause between two words or groups. Normal Morse is 7 dits. You can set 6 to 105.

The pause between the elements (dits and dahs) *inside* a character is always 1 dit. So the characters themselves always come at the set speed, and only the pauses between them get longer. This is the **Farnsworth method**. You learn the sound of a character at full speed and still have time to think.

Next to each spacing slider the app also shows the pause **in seconds** at the current speed, for example “28 dits · 1.68 s @ 20 WPM”.

The **Listen** and **Send** trainings start with generous pauses of **28 / 40 dits**. The CW Keyer, WiFi Trx and the QSO Bot use the normal 7 dits as word spacing.

3.3 Effective speed

Because the pauses are longer, whole words arrive more slowly. The app shows this as **eff.** (effective WPM):

eff. WPM = 50 × WPM / (31 + 4 × character spacing + word spacing)

Example: 20 WPM with 28/40 dits gives 50 × 20 / (31 + 112 + 40) ≈ 5 WPM. The characters sound like 20 WPM, but you have as much time as at 5 WPM. With 3/7 dits the two values are the same.

When copying on the **on-screen keyboard** (Typing) the app waits for you after every word, so the word spacing does not matter. Then only the character spacing counts; the word spacing is taken at the normal 7 dits:

eff. WPM = 50 × WPM / (31 + 4 × character spacing + 7)

With character spacing 3 the value equals the WPM. On the start view, where you have not chosen Paper or Typing yet, the display calculates as for Paper.

3.4 The Koch method

With the Koch method you start with **two characters** at full speed. Once you recognise them reliably, the next character is added, then the next, until you know them all. Each step is called a **lesson**. The lesson number is the number of active characters.

The order of the characters is set by the **Koch sequence** (see Koch sequence).

When drawing random characters, the app weights them like the Morserino. Two out of three characters are drawn evenly from **all** active characters. Every third one is drawn only from the **last third**, where the most recently learned characters are. New characters therefore come up more often, without the old ones disappearing.

3.5 Prosigns

Prosigns (procedural signals) are sent as run-together letters. The app writes them in angle brackets:

Prosign	Meaning
`<ka>`	Start of message
`<ar>`	End of message (+)
`<kn>`	Only the station called should answer
`<sk>`	End of contact
`<as>`	Wait
`<ve>`	Understood
`<bk>`	Break, over to the other station
`<err>`	Error: eight dits (`........`)

In Listen (block flow), a prosign from the groups (**Group characters**, Koch lesson, practice set) sounds as one character and is shown and graded as one character (<ka>). When copying on the on-screen keyboard you type it with the prosign key (AS, KA, KN, SK, VE, BK; it appears as soon as the character set contains prosigns); the two letters K and A count as wrong. The replay “Listen to how it sounds” plays it as a prosign too.

4 Getting started: learning Morse step by step

This chapter is a practice plan for beginners. It does not explain what every setting means (the chapters on each mode do that), but **in which order to do what** and how to set up the app for it. The steps build on each other. You don't have to read everything in one day, and certainly not practice everything in one day.

The basic ideas

- **Listening comes before sending.** If you recognize a character reliably, sending it later is much easier. The other way round works poorly.
- **Learn characters as a sound, not as dots and dashes.** An "A" is not "dot, dash" but a small pattern you recognize as a whole, like a word. If you start counting dots, you will be held back at higher speeds. That is why the app plays characters at **full speed** from the start and only lengthens the pauses (Farnsworth method, see Spacing in dits).
- **Few characters, then slowly more.** You start with two characters. A new one only comes when the previous ones are solid (Koch method, see The Koch method). The app checks this for you.
- **Short and often instead of long and rarely.** The brain learns sound patterns best in small portions with sleep in between.
- **It is not a sprint but a long road.** It takes **several months** until you can copy Morse reasonably well by ear. Until you can also **send**, and at a sensible speed, most learners say it takes about **two years**. These are no guarantees; everyone learns differently and puts in different amounts of time. But don't plan in weeks. There are phases in which nothing seems to improve. That is normal. Enjoy small milestones, such as the next character or the first word you recognize.

How to practice well

- **Several short sessions beat one long one.** About 10 to 15 minutes, a few blocks, two or three times a day if you like. If you stop while it still goes well, you look forward to the next time.
- **Focused, not on the side.** Headphones, quiet, paper and pencil.
- **Breaks are part of learning.** When your hit rate drops, a break usually helps more than carrying on. The app suggests one by itself (see Break hint).
- **The daily goal helps you stay on track.** It counts active practice time; the default is 10 minutes a day. A short day with the goal reached beats a missed one (see Daily goal and streak).
- **Mistakes are part of it.** The app does not treat them as failure; it needs them to give you the right characters more often.

4.1 Step 1: Hear characters (Listen, Koch lesson, Random)

Open **Listen** and set:

- **Character set: Koch lesson**, **Content: Random**.
- **KOCH: 2**. These are the first two characters of the Koch sequence (with **M32**: `m` and `k`). The default is 5; moving the slider back is a good idea for the start. Which sequence you use is secondary; the main thing is to stick to it (see Koch sequence).
- **WPM (character speed): 20**, the default. It sounds fast at first but is right, because characters should arrive as a sound. If you can't get used to it, go to 15. Don't play the characters much slower than that, or you will start counting dots and dashes.
- **Spacing: 28 / 40 dits**, the default. At 20 WPM this gives an effective speed of about 5 WPM (see Effective speed). The app shows it in brackets behind the speed, for example "20 WPM (eff. 5)". If that is too hard, you can widen the spacing further in the ✲ sheet (up to 45 / 105 dits).
- In the ✲ sheet under **Word selection** you can set **Group length** to 3–3 if groups of five are too long at the start.

Before you start, **tap each character of the lesson**: the app plays it three times and shows the code. You don't have to memorize it. Just listen to the sound.

Then press **Paper**. The app plays a block, you copy on paper, at the end it reveals the text, and you mark what was wrong (see How a block runs). Mark honestly: only then can the app judge your progress correctly.

Tip for the first lessons: In the ✲ sheet, switch on **Flow → Stop after each group**. Then you can hear each group again with **Dit** or **REPEAT** until you have it.

4.1.1 When it moves on

You do **not have to raise the lesson yourself**. As soon as every character of the lesson has at least 20 attempts and was recently 90 % correct, the app suggests the next character (see When the next Koch character comes). You accept it or stay a while longer. If you are not sure yet, stay. You can set the thresholds higher or lower in the ✲ sheet under **Adaptive mode**.

The Character statistics show which characters aren't solid yet. If you keep confusing two characters, tap them one after the other and compare them directly.

In this phase the app does **not** shorten the pauses and does not raise the speed (see Pauses and speed). You are meant to focus on new characters without everything getting faster at the same time.

4.2 Step 2: Words, abbreviations and sentences

After a few lessons there are enough characters for real words. Then set **Content** to **Words**, **Abbreviations** or **Mixed**. With the Koch lesson only words made entirely of characters you already know are used; at the start that is very few, with more every lesson. Feel free to switch between

Random and **Words**: random groups train the sound of each character, words train recognizing the whole.

If you catch yourself recognizing a word **as a word** while listening, rather than letter by letter, you are on the right track.

After that, move towards sentences and texts:

- **Head copy** (on the home screen under **Practice**): three listening games where you **write nothing down** but answer questions. This trains understanding Morse, not just decoding it. You don't need a key for them:
  - **Sentences**: you hear one to three short sentences and answer questions about them (Who? What? Where? When?).
  - **Q-groups**: you hear a Q-group (QTH, QRZ, QRM …) and pick what it means. A cheat sheet helps you get started.
  - **Mini QSO**: you hear a short radio exchange between two stations and answer questions about call signs, names, place and report. This is the best preparation for real on-air contacts.

  Each has three levels, from an easy start to the full exchange. Try them as soon as you know most letters and digits.
- **Own texts** (under **Free modes**): copy a text that interests you and listen to it as Morse. With shrinking pauses this becomes a good everyday exercise. The text keeps playing with the screen off, and you can control it from the lock screen or with Bluetooth headphones.
- With **All characters** and **Callsigns** you practice how callsigns sound. This is especially useful for radio amateurs.

4.3 Step 3: Send yourself

As soon as you hear a few characters reliably, you can start **Send** (the Echo Trainer). The app plays a word or group and you send it back with the key. Here you have a **profile of your own**, with its own Koch lesson, its own speed and its own statistics, separate from **Listen**. Start with few characters, even if you are further along when listening.

Settings for the start:

- **Listen** (speed of the prompt): as in Listen.
- **Send** (speed of your answer): better a little lower than the listening speed (not with the straight key, where the app measures your speed). Your sending should be **even**, not fast. Clean slow sending beats fast bumpy sending (see Sending speed).
- **Prompt = Display** helps to get from the written word to sending: you see the word and send it without hearing it.
- You practice a single new character by **long-pressing** it: the app plays it again and again and you send it back (see Practicing a single character). It doesn't count in any statistic, so it is stress-free.

4.3.1 The key: optimum and alternative

A real Morse key is the optimum. A phone has no input for it, so you need a small **USB adapter** that turns the key contacts into key presses, for example the vband or a homemade one. Which adapters exist and how to connect and teach it is in the chapter A real Morse key or straight key. You feel the pressure point, the movement is the same as later at the radio, and you don't have to look at the screen. Which key you take depends on the goal:

- **Paddle (dit/dah)** with keyer mode **Iambic A** or **B**: the standard on the air; the keyer produces the characters evenly.
- **Straight key** with keyer mode **Straight**: the app measures the speed from your keying (see Straight key: automatic speed). If you send much slower than 15 WPM, set **Straight key start speed** to your speed.

The on-screen touch keys are a good alternative for the start and on the go, and for Send, the games and the CW Keyer they are **fully sufficient up to a point**. They have limits, though:

- No pressure point; you have to look, and you don't hit the areas equally well every time. This slows you down, above all at higher speeds.
- Iambic squeeze (both keys at once) is awkward on a touchscreen.
- The movements you get used to are different from those on a real key.

If you want to continue seriously with keying, plan for a real key sooner or later.

4.4 Step 4: Shorten pauses, raise speed

So far the effective speed (about 5 WPM) was much lower than the character speed (20 WPM). That changes now, in this order:

1. **Shorten the pauses.** As soon as all characters of the Koch sequence are unlocked (or you practice with **All characters**), the app suggests shorter pauses after two good blocks in a row, one dit per step, down to the normal 3 / 7 dits (see Pauses and speed). You can also do this yourself: with **Character spacing** and **Word spacing** in the card below the practice area.
2. **Only then raise the speed.** Once the pauses are at 3 / 7, the app suggests +1 WPM when things go well. Small steps, no jumps.
3. **If it goes badly, it goes back.** If the block EMA falls below the lower threshold (70 %), the app suggests longer pauses. That is not a punishment but the right move: better one step back than skipping everything.

All suggestions appear on the result page. You decide whether to accept them (see Adaptive mode).

4.5 Step 5: Add interference

Under real conditions signals are rarely clean. With **Settings → Interference** you simulate noise, fading, a second station and a “bad fist” (see Interference). Start carefully, and only once you are reliable at normal pauses:

1. **Light**: little noise and slight fading.
2. **HF evening**: noticeable noise, fading and a neighbouring station.
3. **Pile-up**: demanding, with a loud second station.

The quickest way to switch interference on and off is the **wave symbol** in the header. With **Constant noise** the noise is also there between characters, which is more realistic but more tiring. Practice with interference best with characters you already know well, not with brand-new ones. Your hit rate will often drop briefly with interference. That is expected.

4.6 Afterwards: practice where it is fun

Once the basics are in place, you can use everything else with a purpose:

- **QSO Bot**: a simulated radio conversation in which you learn callsigns, reports and phrases.
- **CW Decoder**, **WiFi Trx**: read CW from a radio or connect with other Morserinos.
- **Games**: variety that still trains listening and sending.
- **Learning resources**: Morse tree and character table for looking things up.

When it doesn’t move on

- **One character holds up the unlock.** The statistics show which one. Tap it, listen to it calmly and keep practicing for a few blocks (the app boosts weak characters by itself).
- **Two characters get confused.** Good that you know: play them one after the other and compare the sound.
- **It feels like a plateau.** That is normal and usually resolves after a few days of break or other exercises. Change the content, practice shorter, take a rest day.
- **You are annoyed.** Then stop for today. Annoyance is a bad teacher.

5 Listen: practice copying

Listen is the app's CW Generator and Koch Trainer. The app plays a **block** of groups or words, and you copy it, in one of two ways:

- **Paper**: you copy on paper. Nothing is shown on screen while it plays. Afterwards the app reveals the text and you tap what you got wrong.
- **Type**: for on the go. The app plays one word after another and you type it on the app's own on-screen keyboard (see Copying on the on-screen keyboard).

Either way you get the same result with suggestions for the next block at the end, and both count towards the same Listen statistics.

5.1 Choosing the character set and content

At the top of the start view you choose two things.

The **character set** decides which characters are practiced:

Character set	Meaning
Koch lesson	Only the characters up to your current lesson
All characters	All letters, digits, punctuation and prosigns
Practice set	Only the characters you enter yourself

The **content** decides what is played:

Content	Meaning	Available with
Random	Groups of random characters	all
Words	Common words (English or German, weighted by frequency). The chip shows roughly how many different words can come up (e.g. "~190", or "none fit" when nothing matches): it depends on the language, the word length and, with Koch, on the unlocked characters	Koch, All
Abbreviations	Common CW abbreviations	Koch, All
Call signs	Random, realistic call signs	All
Mixed	With Koch: words, abbreviations and groups mixed. With all characters: words, abbreviations, call signs and single punctuation marks or prosigns, as on the Morserino	Koch, All

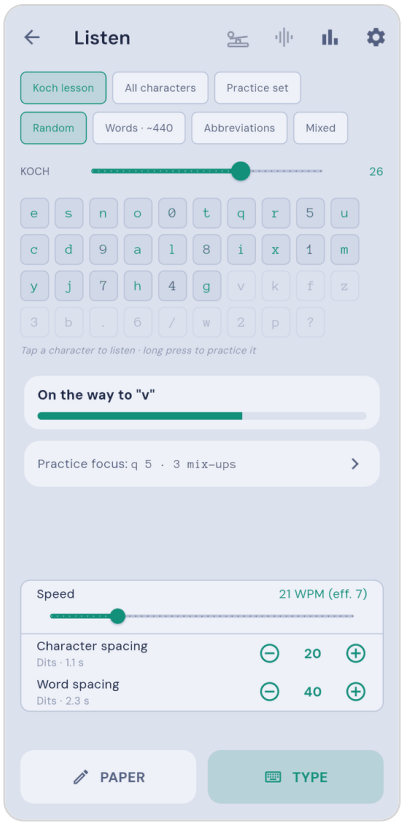
With the Koch lesson, only words and abbreviations made **entirely** of characters you have already learned are used. In the first lessons there are very few of them.

5.1.1 Setting the Koch lesson

The **KOCH** slider sets the lesson. Below it you see every character of the Koch sequence. The characters active in this lesson are highlighted, and their color shows the kind of character: letters, digits and punctuation/prosigns are colored differently. Characters not yet unlocked by the lesson are dimmed.

To get to know a character (this works for the dimmed ones too, so you can listen ahead to later lessons):

- **Tap** it to play it three times at the current speed. A tile with the character and its Morse code appears in the middle of the screen, and the rest of the screen is dimmed. The dits and dahs start out gray and light up exactly while they sound. Each repetition starts from gray again. After the third repetition the tile closes by itself. A tap anywhere on the screen (or the back button) stops playback at once.
- **Long press** it to open the practice page for that character: listen and, if you like, key it back (see Practicing a single character).

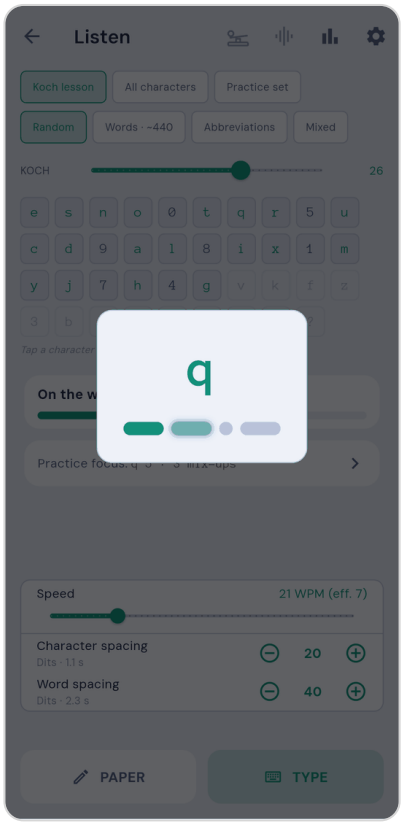


Listen start view: character set, content, Koch lesson, weak characters, spacing and speed

Usually you don't need to raise the lesson by hand. The block flow suggests the next character once you are ready (see When the next Koch character comes).

5.1.2 Practice set

With the **Practice set** character set, an input field appears right on the start view. Enter the characters you want to practice, for example `QXZJ`. Upper/lower case and spaces don't matter, and duplicate characters are ignored. The app shows how many distinct characters it found. The practice set only offers the **Random** content.



Tapping a Koch character: tile with its Morse code

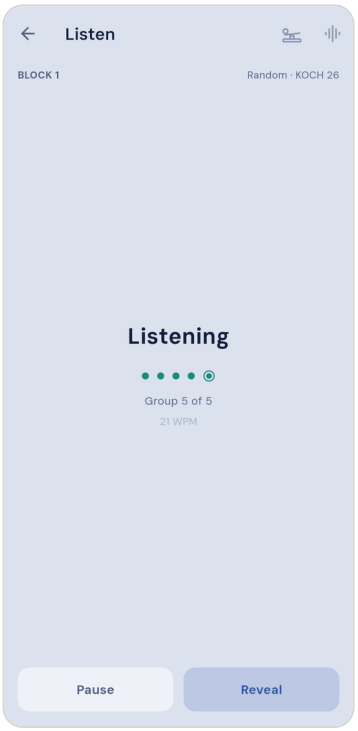
5.2 Speed

Below the practice area is a card with **Speed** (character speed, 10 to 60 WPM, the effective speed in brackets behind it), **Character spacing** and **Word spacing**. It is the same as in Sentences, Mini QSO and Q-groups. You see it on the start view and in the result, not while a block runs and not in "Sent" (nor while you mark single words). You can also set the pauses in the ⚙ sheet under **Spacing**.

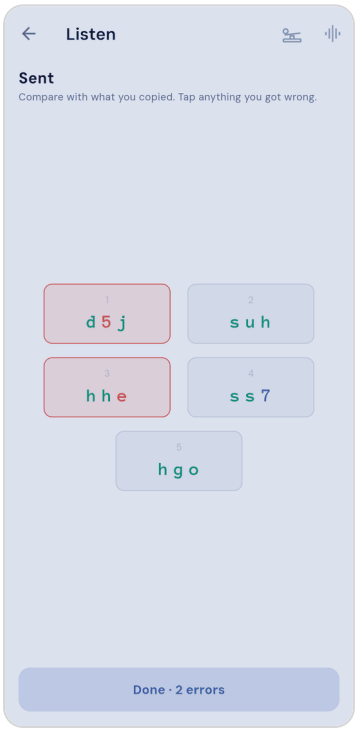
5.3 How a block runs

At the bottom of the start view are two start buttons: **Paper** and **Type**. The one you used last is highlighted. This section describes **Paper**; **Type** follows further down.

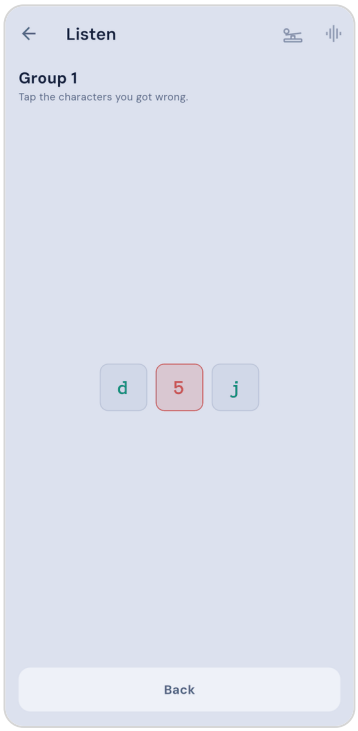
1. Press **Paper**. After one second of "Get ready …" the block begins.
2. The app plays the groups one after another, with the set word spacing in between. Only "Group *n* of *N*" (for words "Word *n* of *N*") and the speed are shown, not the text. Copy on paper.
    - **Pause** stops after the current group, and **Resume** continues.
    - **Reveal** ends the block early and shows only the groups played so far.
3. After the last group, **Sent** appears, with each group as its own tile. Compare it with what you copied.
4. **Marking errors:** tap a group where you made a mistake. It opens large, one character at a time. Tap every character you got wrong or missed, then **Back**. Tapping a character again removes the mark. The marked characters show in red in the overview.
5. **Done · *n* errors** finishes the evaluation. The results are saved now, and the result page appears.



During the block: progress only, no text



Revealed: groups with errors are red



One group opened: tap the wrong characters

You set how many groups a block has in the ⚙ sheet under **Word selection → Groups per block** (with words, the setting is called **Words per block**). During the block and on the result page the app says **groups** for Random and **words** for all other contents.

5.3.1 Stop after each group

With **Flow → Stop after each group** (in the ⚙ sheet; with words it is called **Stop after each word**), the app waits after every group or word:

- **Dit** (left key) or the **REPEAT** button plays the same group again.
- **Dah** (right key) or **NEXT** plays the next group.

This matches “Stop<Next>Rep” on the Morserino. It’s handy at the start, when you want to hear a group several times. The setting only applies to **Paper**; when typing, the app waits after every word anyway.

5.4 Copying on the on-screen keyboard

With **Type** you copy on the app’s own keyboard on the display instead of on paper, meant for on the go. It is not the phone’s system keyboard.

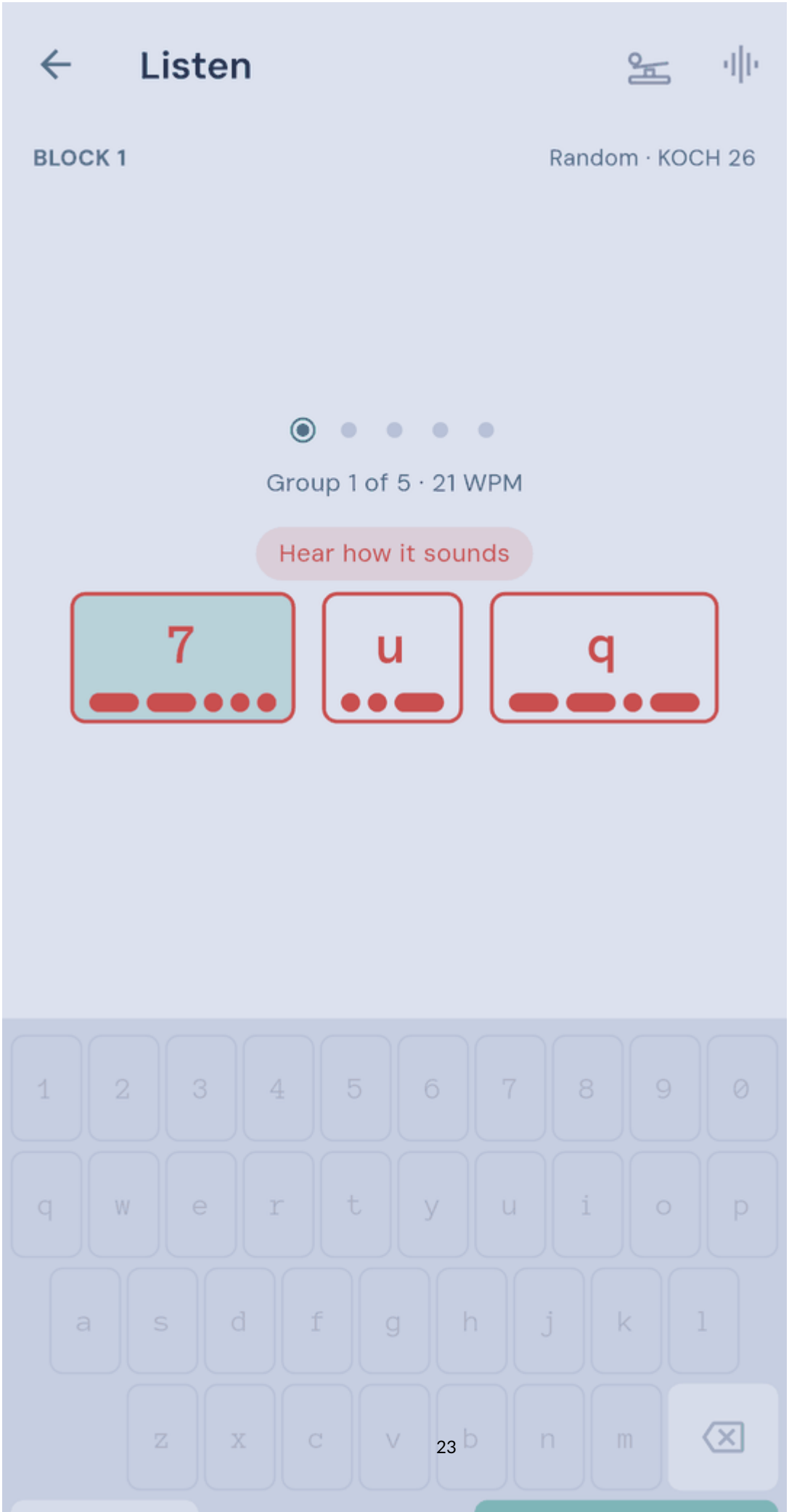
The keyboard is a QWERTY keyboard with a digit row above it. Every key is always in the same place. Only the characters the chosen character set can contain are active (bright, tappable); the others are only outlined and do nothing. If the character set contains punctuation (`. , : – / = ? @ +`), it gets a row of its own. While you press a key, a bubble above it shows the character large. ⌫

deletes the last character. **Pass** is at the bottom left, ↵ **Check** at the bottom right. While the block runs, the WPM slider is hidden to make room for the keyboard.

How a word runs:

1. Press **Type**. After one second of “Get ready …” the app plays the first group (or word). “playing …” is shown below the answer line.
2. You can type along while it plays or wait until it has finished; both work. There is no time limit.
3. **Checking:** as soon as you have typed as many characters as the word has, the app checks automatically (after the word has ended plus a short 0.4 s, during which you can still correct with ⌫). ↵ **Check** hands in earlier, for example when you missed a character. If you press ↵ while the word is still playing, the app checks right after the word ends.
4. ✓ **Correct**: the next word comes after about one second. Whatever you type during that second counts for the next word.
5. ✗ **Wrong**: the word is played again at once, with an empty field. Your previous attempt is shown small and struck through above it, without a hint where the error was. “Attempt *n* of *max*” shows which attempt this is. With the setting **After a wrong answer → Learn** (see below) there is no second attempt: the word sounds once more, each character is shown below as dots and dashes (the one sounding lights up), then the next word follows. Only the first attempt is still graded; a **Pass** also shows the lesson (with the error tone, all characters red). Wrong characters are red and also framed, so they stand out in the Red theme too.
6. **Pass**: at any time, also while the word plays. After the last wrong attempt or a pass, the app shows the solution for 2 seconds (the characters wrong in the first attempt in red) with your attempts below it, then moves on.

::: {.shots one}


Listen
BLOCK 1
Random · KOCH 26
Group 1 of 5 · 21 WPM
Hear how it sounds
7
u
q



Typing: the keyboard, only the lesson's characters are active



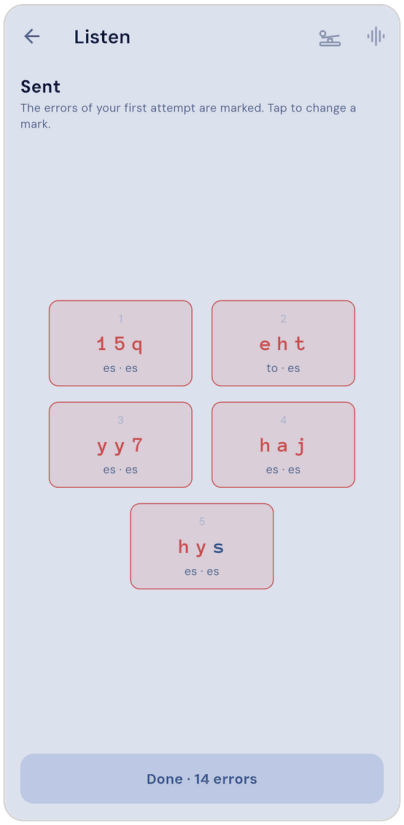
Wrong: the word plays again, attempt 2 of 2

The confirmation tone (high = correct, low = wrong) follows the **Confirmation tone** setting in the ⚙ sheet of **Send**. How many attempts you get per word and whether the keys vibrate is set in the ⚙ sheet under **Flow** (see below).

After the block, **Sent** appears as with Paper. The errors of your **first** attempt are already marked red, and under each group is what you typed ("— passed" for a pass). You can tap a group and change marks, for example when you only mistyped. **Done · *n* errors** saves and shows the usual result page.

How it is counted:

- Only the **first** attempt of each word counts; the second is easier because you hear the word a second time.
- The app compares character by character, but not rigidly position by position: if you left out a character, only that character is wrong, not every one after it. Example: played `tqr5u`, typed `tq5u` → only `r` is wrong. A character heard wrong is wrong (`cd9al` as `cb9al` → `d` wrong). An extra typed character does not make any played character wrong.
- A key pressed twice at the very end (`ABCC` for `ABC`) counts as a typing slip: the word is ✓ right and not repeated. Any other extra character (`ABCE`, `AABC`) is still wrong, and the word is repeated.
- **Pass** in the first attempt means every character of that word is wrong, like a blank on paper.
- When typing, the pauses between words don't matter. The app therefore only suggests changes to the **character spacing** (see Pauses and speed).



Sent after a typed block: first-attempt errors marked, your input below

The arrow at the top left cancels the block; nothing is saved then. The keyboard is portrait-only for now.

5.5 The result page

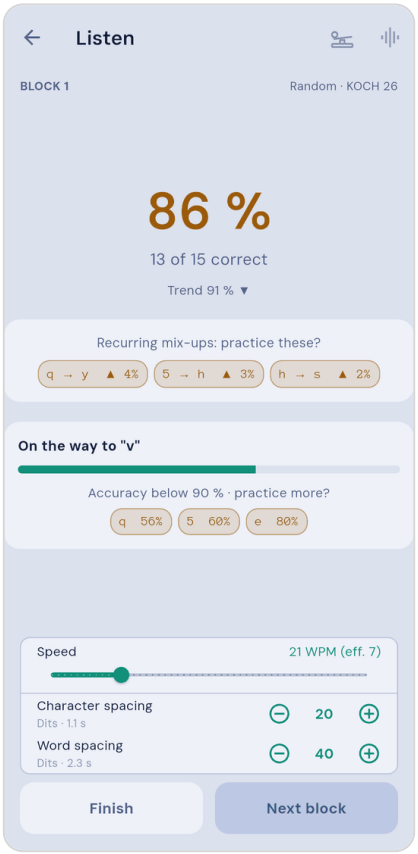
From top to bottom, the result page shows:

- **Accuracy** in percent, meaning the share of characters copied correctly, plus "*x* of *y* correct". It is green from 90 %, yellow from 70 % and red below that.
- The **trend** line, from the sixth block on (see trend). Speed, effective speed and spacing are in the card at the bottom, above the buttons (the same on the start view).
- The **speed and spacing card** (not in the result list): − and + change **Character spacing** and **Word spacing** by 1 dit each, the slider the speed. This takes effect immediately and doesn't depend on the suggestions. Once you started **Typing**, the word spacing is missing, because it has no effect there.
- **Weak characters:** the characters that cause you the most errors over time, with their error rate (see Weak characters).
- **On the way to "X"** (Koch lesson only): what you still need before the next character (see Progress card).
- **Suggestions:** what the app recommends for the next block (see Accepting, rejecting, adjusting suggestions).

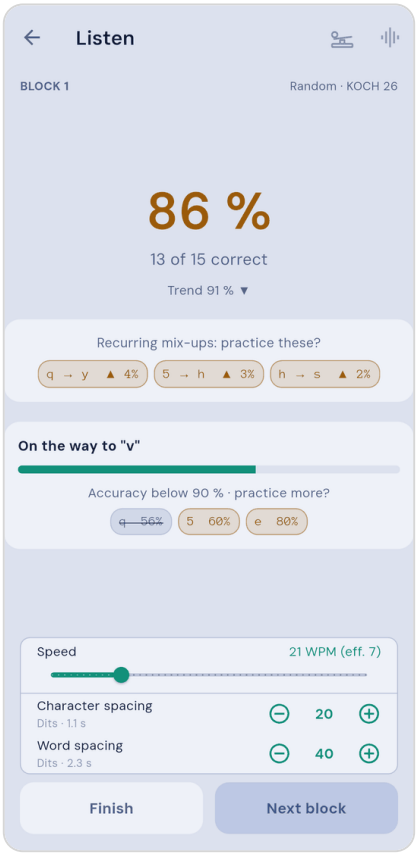
At the bottom there are two buttons:

- **Next block** applies the ticked suggestions and starts the next block right away.
- **Finish** also applies the ticked suggestions but doesn't start a new block.

How the app arrives at its suggestions is explained in the chapter Adaptive mode.



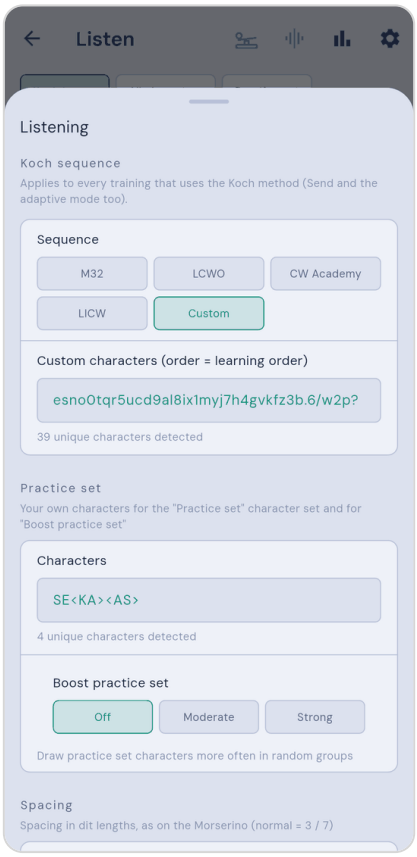
Result page with accuracy, spacing, weak characters and progress card



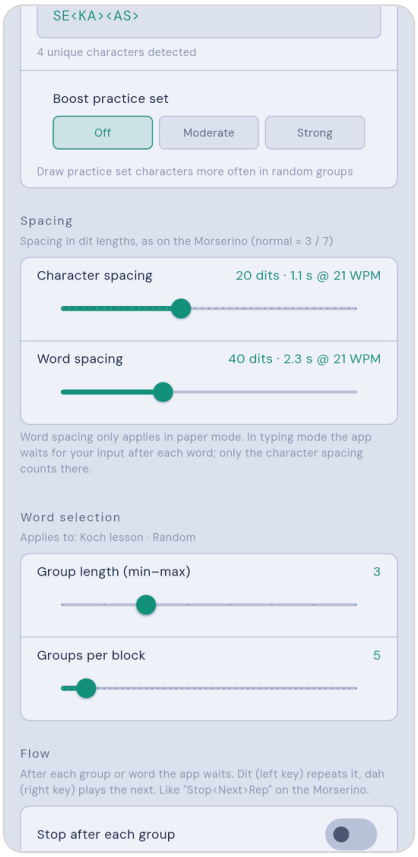
A weak character excluded (struck through)

5.6 Settings in the ⚙ sheet (Listen)

The ⚙ sheet of **Listen** has the following sections. Values in **bold** are the defaults.



⚙ sheet: Koch sequence and practice set



⚙ sheet: Spacing, Word selection, Flow

5.6.1 Koch sequence

Only shown when the **Koch lesson** character set is selected. This setting applies to **all** trainings and games that use the Koch method. For a description see Koch sequence.

5.6.2 Practice set

Setting	Meaning	Values
Characters	The practice set characters. This is the same field as on the start view with the **Practice set** character set	any characters
Boost practice set	Draws practice set characters more often in random groups (see below)	**Off** / Moderate / Strong

This is how **Boost practice set** works. For each character of a random group, the app draws up to 3 times (Moderate) or 8 times (Strong) until it gets a character from the practice set. If that doesn't happen, the last character drawn stays. The practice set characters come up more often, but the others don't disappear.

Boost practice set works in **Listen** (Koch lesson or All characters · Random) and in **Send** (All characters · Random). In **Listen** the practice set is combined with the weak characters: the characters from both lists are boosted, at the higher of the two levels. The more characters come together, the less each single one stands out.

5.6.3 Spacing

Setting	Meaning	Values
Character spacing	Pause between characters, in dits	3–45 (**28**)
Word spacing	Pause between groups/words, in dits	6–105 (**40**)

Word spacing can never be smaller than character spacing. If you move character spacing past it, word spacing is pulled along.

Word spacing only applies in paper mode. In typing mode the app waits for your input after each word, so only the character spacing counts there. The ✲ sheet notes this below the sliders.

5.6.4 Word selection

Only the settings that fit the chosen content are shown. The line “Applies to: …” tells you which combination you are setting up.

Setting	Meaning	Values
Group characters	Only with **All characters · Random**: which character classes are drawn from	**All** / Letters / Digits / Punctuation / Prosigns / Letters+Digits / Digits+Punct. / Punct.+Prosigns / Letters+Digits+Punct. / Digits+Punct.+Prosigns
Group length (min–max)	Characters per random group (with **Random** and **Mixed**). A range slider: with min = max (e.g. 5–5) all groups are equally long; with min < max (e.g. 2–7) each group gets a random length, so you can’t tell while listening whether a group is over	1–8 (**5–5**); with 1–1 you practise single characters
Word language	Which word list is drawn from (with **Words** and **Mixed**): English or German. Both lists are weighted by frequency: common words come up more often than rare ones. The German list has no umlauts and no ß	**English** / German

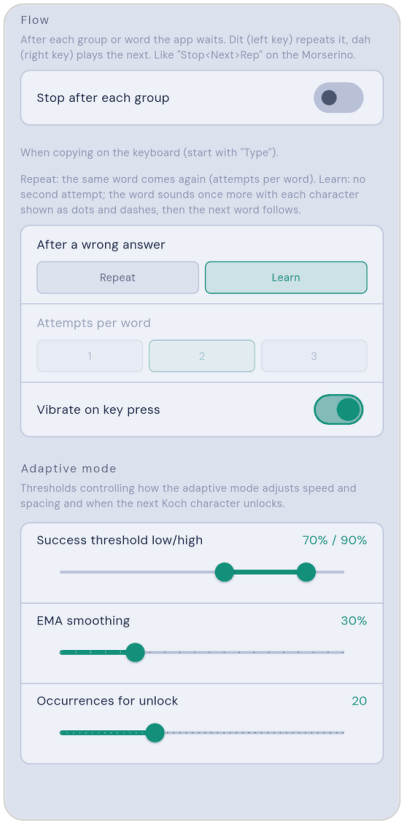
Setting	Meaning	Values
Word length (min–max)	Only words within this length range (with **Words** and **Mixed**); the open ends mean no limit	1–**all** (1–8, **all**)
Abbreviation length (min–max)	Only abbreviations within this length range (with **Abbreviations** and **Mixed**); the open ends mean no limit	2–**all** (2–6, **all**)
Max call sign length	Only with **Call signs**, and with **Mixed** (All characters): maximum length of the call signs. One value for Listen and Send	**Unlim.** / 3 / 4 / 5 / 6
Region	Same, only call signs from this region	**All** / EU / NA / SA / AF / AS / OC / VK/ZL
Common prefixes only	Same, only frequently heard prefixes instead of all possible ones. Call signs follow a weighted prefix table, as on the Morserino	Off / **On**
Groups per block / Words per block	Number of groups (with **Random**) or words in a block	1–50 (**10**)

5.6.5 Flow

Setting	Meaning	Values
Stop after each group / Stop after each word	**Paper** only: wait after each one, dit = repeat, dah = next	**Off** / On
After a wrong answer	**Type** only: **Repeat** = the same word comes again (attempts per word); **Learn** = no second attempt; instead the word sounds once more and shows each character as dots and dashes with the letter above it, then the next word follows	**Repeat** / Learn
Attempts per word	**Type** and **Repeat** only: how often you may try a word; 1 = no second attempt	1 / **2** / 3
Vibrate on key press	**Type** only: short vibration on every active key	Off / **On**

5.6.6 Adaptive mode

These are the thresholds the app bases its suggestions on. They apply to **Listen and Send** together and are described in detail in Adaptive mode settings.



⚙ sheet: After a wrong answer, Adaptive mode

6 Send: Echo Trainer

Send is the Echo Trainer. The app plays a word or group, and you key it back with the Morse key. That can be the touch keyer or a real Morse key (see Morse key). If you get it right, the next word comes. If not, it is repeated. Keying with a real Morse key on an adapter feels much better than with the touch keys; the first time you open Send, a short tip says so (close it with the ×, it does not come back).

Send has its **own profile**, independent of Listen. It keeps its own Koch lesson, speed, spacing, practice set and character statistics. What you mix up when listening isn't necessarily what you get wrong when sending.

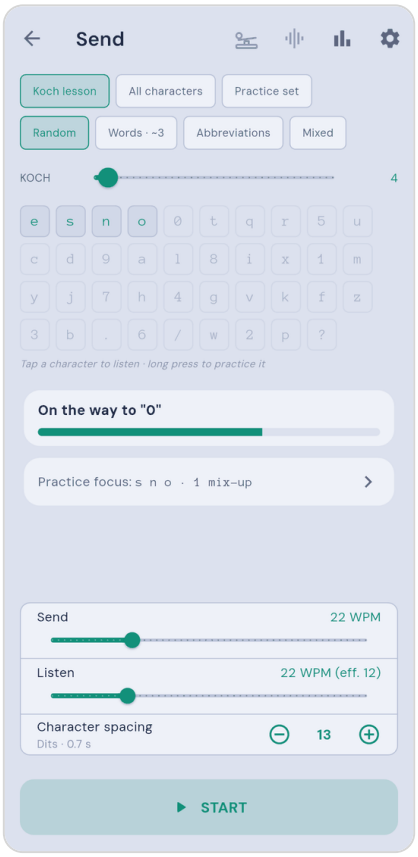
6.1 Start view

At the top you choose the **character set** and **content** as in Listen (see Choosing the character set and content). With the Koch lesson you also choose the lesson. Tapping a character works as in Listen.

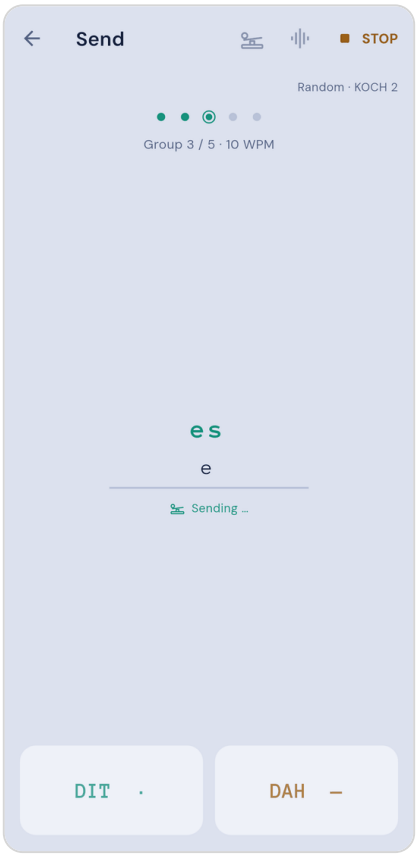
Below that, two sliders set the speeds:

- **Listen** is the speed at which the word is played to you (10 to 60 WPM).
- **Send** is the highest speed at which your answer is expected (see Sending speed). "same as prompt" means the same speed.

At the bottom are the dit/dah keys or the straight key, and **Start**.



Send start view



While answering: prompt (Prompt = Both), attempt and speed

On the **start view** the same **speed and spacing** card sits at the bottom as on the result page. The **Send** speed is at the top (the leftmost step is “same as Listen”; with a straight key the slider only shows the measured speed), below it the **Listen** speed with the spacing. The touch paddles only appear once the block runs, and then sit at the very bottom like in the other keying screens. You end the block with **Stop** at the top right or with the back arrow.

6.2 How a word runs

1. After **Start** the app waits 2 seconds, then plays the first word.
2. **Your answer:** key the word back. Your sidetone is shifted by half a tone so you can tell prompt and answer apart. This is the **Tone shift** setting.
3. **When you have to start:** from the end of the prompt you have about 1.4 seconds, plus one character pause, plus a third of the **answer pause**, plus the **Think time** (default 8 s) to **begin** your answer. If nothing comes by then, the word counts as wrong.
4. **When the answer ends:** once you have started, the think time no longer applies. The answer is complete as soon as you pause long enough after a character (formula under Settings in the ⚙ sheet), at sending speed.
5. **Correcting:** key `<err>` (eight dits) or four `e` in a row to clear your answer so far and start again. The exception is when the word itself continues with another `e` at that point. Then the `e` counts as a normal character.

6. **Grading:**
    - ✓ **Correct**, optionally with a confirmation tone. The next word comes after a little more than a second.
    - ✗ **Wrong**: the word is played again. "Attempt *n* of *max*" shows which attempt this is. Once all **Repeats** are used up, the app shows the correct word for two seconds and moves on.

Whether the prompt is played, shown or both is set with **Prompt** (see below).

6.3 Sending speed

On the Morserino this setting is called "Echo Speed Max". It limits the speed at which **your answer** is expected. The prompt still plays at the listening speed.

Example: Listen at 25 WPM and Send at 18 WPM means you hear fast but may answer more slowly. The sending speed sets how fast the keyer produces your dits and dahs and how long a word pause has to be. At the far left ("same as prompt"), the listening speed applies to the answer too. The lowest sending speed is 10 WPM.

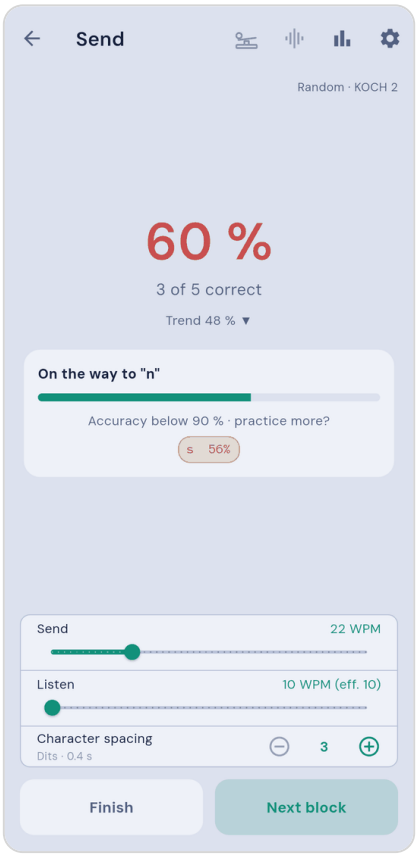
With the **straight key** (keyer mode Straight) there is no set sending speed: the app measures it from your keying (see below). The **Send** row is then disabled and shows the measured speed.

6.4 The result page

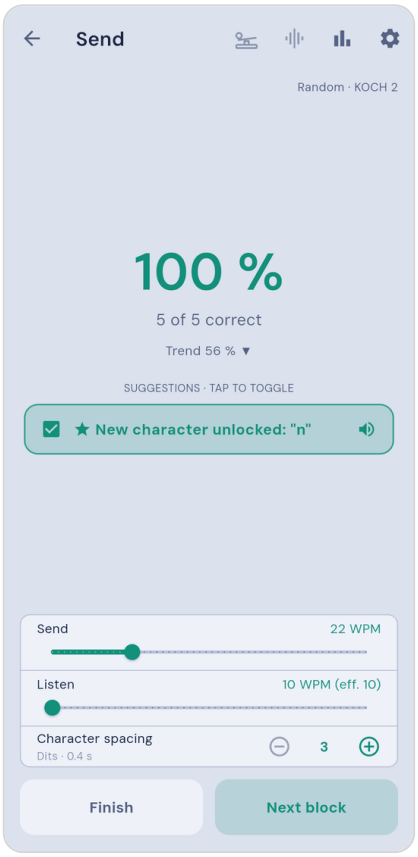
After the last word of a block the result page appears:

- **Accuracy** is the share of words correct on the **first attempt**. The colours are the same as in Listen. Below it come "x of y correct" and the trend. As in Listen, there is no list of the single words.
- The **speed and spacing** card (at the bottom, above the buttons): sliders **Send** and **Listen** (with the effective speed in brackets), plus **Character spacing**. The answer pause is not on this screen but in the ⚙ sheet; the effective speed counts it at the normal 7. The spacing belongs to the Listen speed directly above it. A change here replaces the matching suggestion.
- **Suggestions** and **Weak characters** (see Adaptive mode).

Next block (right) applies the ticked suggestions, boosts the ticked weak characters in the next block, and starts it. **Finish** applies the ticked suggestions without a boost and returns to the start view.



Result page: accuracy, progress, weak characters



A suggestion (here: new character unlocked), ticked

6.5 Settings in the ⚙ sheet (Send)

The **Koch sequence**, **Practice set**, **Spacing** and **Word selection** sections are the same as in Listen (see Settings in the ⚙ sheet (Listen)), but they apply to the Send profile. There are two differences:

- **Spacing** applies to the word played to you **and to your answer**, as on the Morserino. Your answer counts as finished once you pause this long after a character:

  ```
  2 × character spacing + 1 + answer pause / 8   (dits at the answer speed)
  ```

  With a straight key it is answer pause + 1 dits, measured at your own speed (up to 30 WPM; above that the shorter straight-key gaps apply). Example: character spacing 28, answer pause 40, answer speed 18 WPM gives 62 dits, about 4 s. That is how long you may pause between the characters of a group, and how long the app waits after the last character before it scores. If that is too lenient or too slow for you, lower the character spacing; the normal 3 / 7 gives 8 dits. When the adaptive feature shortens the character spacing, the answer gets stricter too. A longer character spacing also gives you more time to begin your answer. The **Spacing** section has only the character spacing here; the answer pause is in the **Echo Trainer** section and does not depend on it.

- **Word selection** shows the same call sign settings as in Listen (one value for both).

In addition there is the **Echo Trainer** section:

Setting	Meaning	Values
Think time	Extra time after the played word to **begin** your answer (for example for head copy). It only applies until your first character	0–20 s (**8 s**)
Answer pause	Silence after your last character before the answer counts (formula above). It is never played and independent of the character spacing. Also extends the time to begin a little (a third of it)	6–105 dits (**40**)
Repeats	How often a wrongly answered word is played again before the app reveals it. “Forever” repeats until you get it right	0–6 (**3**), Forever
Prompt	How the prompt is given. **Sound** means you only hear it. **Display** means you only read it, with no audio. **Both** means you hear it and then read it once it has played	**Sound** / Display / Both
Sending speed (max)	Highest speed for your answer, see Sending speed	**same as prompt**, 10–50 WPM
Tone shift	Your sidetone while answering is half a tone above or below the prompt	No shift / **Up ½** / Down ½
Confirmation tone	Short tone after grading: high for right, low for wrong	Off / **On**

Prompt = Display is a good way to get from written text to sending, for example to drill new characters.

6.6 Practicing a single character

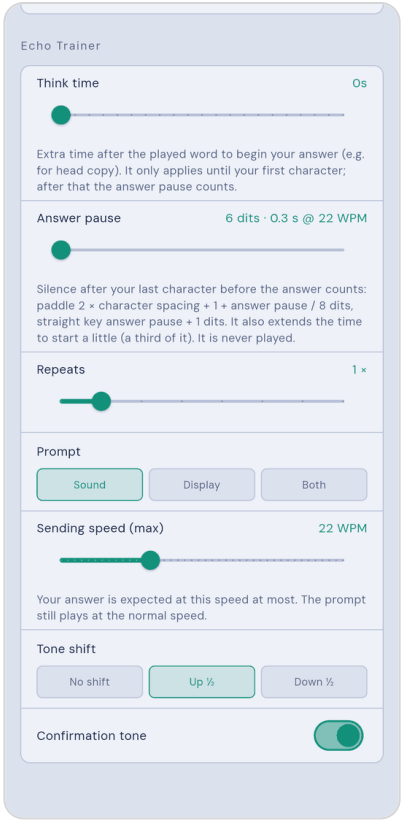
In Listen or Send, long press a Koch character (locked ones too) to open **Practice: X**. In the middle is the same tile as for a tap: the character and its Morse code, whose dits and dahs light up while it plays. The character plays over and over.

After each play you **can** key the character back, with the touch keyer at the bottom or a connected key, but you don't have to. Your dits and dahs appear below the line in the tile as you key them. Once the character is complete they turn green (✓ **Correct**) or red (✗ **You keyed:** with the character you keyed), and shortly after the character plays again. If you key nothing, it repeats after a pause. That is no error, and nothing is counted or added to any statistics. **Back** leaves the page.

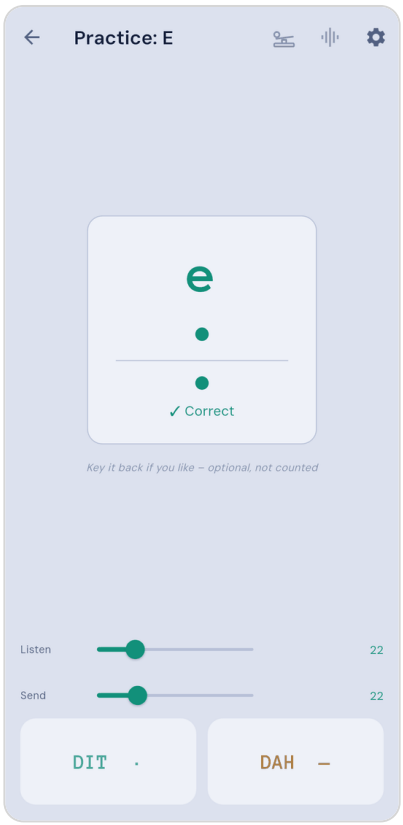
Above the keyer two sliders set the speed: **Listen** is the speed the character plays at, **Send** the speed you key back at (the leftmost step is "same as prompt"; with a straight key it only shows the measured speed). They are the same two settings as in Send, so a change applies there too, from the next play on.

Set the length of the pause with the **gear** icon at the top right: **Pause before repeating**, 1–20 s (default **4 s**). It applies only to this practice page, not to Send.

This corresponds to "Learn New Chr" or "Preview Char" on the Morserino.



Send ⚙ sheet: the Echo Trainer section



Practice: a character, keyed back correctly

7 Adaptive mode

Listen and Send always run in **blocks**. After each block the app evaluates how it went and gives you **suggestions**. It may suggest unlocking the next Koch character, making the pauses shorter or longer, or raising the speed. **None of this happens behind your back.** Every suggestion appears on the result page, and you decide whether to accept it.

This chapter explains exactly how the app calculates. You don't need it to practice, but it helps you understand the suggestions and set the thresholds deliberately.

7.1 What the app tracks

7.1.1 Per character

The app keeps statistics for every character, separately for **Listen** and **Send**:

- **attempts**: how often the character was graded,
- **errors**: how many of those were wrong,
- **moving error rate**: the basis for all decisions.

The **moving error rate** is an exponential moving average (EMA). On every attempt:

new rate = 0.2 × (1 on error, else 0) + 0.8 × old rate

Each new attempt counts 20 % and the past counts 80 %. Recent results weigh more than old ones, but a single slip doesn't upset everything. For example, a character at 0 % error rate that gets one error is at 20 %. It then takes 3 correct attempts to get back below 12 %, and 8 correct attempts to get below 5 %.

A character's **accuracy** is 100 % minus its moving error rate.

What counts as an attempt?

- **Listen:** every character played in a block. It is correct unless you marked it as an error. When copying on the on-screen keyboard, the app marks the errors of the **first** attempt of each word itself (see Copying on the on-screen keyboard).
- **Send:** only the **first** attempt of each word. The characters before the first error count as correct, and the first wrong character counts as an error. The characters after it aren't counted, because it's unclear whether you heard them correctly. Repeats of the same word don't count.

7.1.2 Per block

Every block has a **block rate**:

- **Listen:** characters copied correctly / all characters.
- **Send:** words correct on the first attempt / all words.

From the block rates the app forms another moving average, the **block EMA**:

block EMA = α × block rate + (1 − α) × previous block EMA

α is the **EMA smoothing** setting (default 30 %). The block EMA starts at 100 %. It is kept across sessions, separately for Listen and Send.

7.2 The two thresholds

With **Success threshold low/high** (default **70 % / 90 %**) the app divides the block EMA into three ranges:

Block EMA	Meaning	Suggestion
90 % and above (high)	Going well	After **2 blocks in a row** in this range: shorten the pauses, or raise the speed if the pauses are already normal
70 % to below 90 %	About right	Change nothing
below 70 % (low)	Too hard	Immediately: lengthen the pauses

The high threshold is also the accuracy every character has to reach before the next Koch character unlocks.

7.3 When the next Koch character comes

The next Koch character is suggested when **every** character of the current lesson meets two conditions. That means all of them, not just the most recently learned one:

1. at least **20 attempts** (the **Occurrences for unlock** setting), and
2. an **accuracy of at least 90 %** (the high success threshold).

Two points matter here:

- The condition applies to each character individually. A single weak or rarely practiced character holds up the unlock. The progress card and the character statistics show you which one it is.
- Because accuracy is a moving value, it isn’t enough to have got a character right 20 times at some point. Your **recent** attempts have to be good.

The unlock appears as a highlighted suggestion with a star: **New character unlocked: “X”**. The 🔊 icon next to it plays the new character three times, without leaving the result page. It shows the same tile as tapping a Koch character: the character with its code below, each element lighting up as it sounds. A tap closes it early. If you untick it, you stay in the current lesson.

7.4 Pauses and speed

The app always changes the **pauses** first, and only then the speed:

- **Shorten:** character spacing and word spacing each get 1 dit shorter, down to the normal 3 / 7 dits. In Send only the character spacing changes; the answer pause stays as you set it.
- **Raise the speed:** only once the pauses are already at 3 / 7 dits does the app suggest **+1 WPM**.
- **Lengthen:** character spacing and word spacing each get 1 dit longer, but never longer than at the start of the session. In Listen that is when you opened the training. In Send it is the value from the ⚙ sheet.

When copying on the **on-screen keyboard**, the app only changes the **character spacing**; the word spacing stays, because the app waits for you after every word there anyway. The pauses then count as normal as soon as the character spacing is at 3 dits. For the same reason the **speed and spacing** card shows only the character spacing once you start **Typing**, and the ⚙ sheet notes at the word spacing that it only applies to **Paper**.

While you are still working through the Koch sequence, the app neither shortens the pauses nor raises the speed. You should be able to concentrate on new characters without everything getting faster at the same time. Lengthening is always possible. Shortening and raising the speed only start once one of these is true:

- all characters of the Koch sequence are unlocked, or
- you practice with **All characters** or a **Practice set**.

The block in which a new character is added also never shortens the pauses or raises the speed.

You can step in yourself at any time, independently of these rules. In Listen use the **speed and spacing** card, and in Send use the speed controls.

7.4.1 Sending speed in Send

Send has one extra suggestion: **Sending speed increased** (+1 WPM). It only appears if you have set a sending speed **below** the listening speed and the block reached at least the high threshold. It starts **unticked**, because the sending speed is a deliberate choice. With the straight key this suggestion is dropped, and so are the tighter/wider spacing suggestions; listening speed and new characters are still suggested.

7.5 Accepting, rejecting, adjusting suggestions

Each suggestion is a row on the result page:

- The **checkbox** on the left accepts or rejects it. Most suggestions start ticked.
- **−** and **+** adjust the size. The speed goes up to at most 5 WPM above the current one. The pauses can be anywhere between 3 / 7 dits and the value at the start of the session.
- The status line immediately shows the values for the next block.

Nothing is applied until you leave the result page with **Next block** or **Finish**.

7.6 Weak characters

A character counts as **weak** if both of these are true:

- it has at least **8 attempts**, and
- its moving error rate blocks the **upper threshold**: at least *100 % minus the upper threshold*, but at most **12 %** and at least **2 %**. With the default threshold of 90 % that is **10 %**, with 95 % it is **5 %**.

So a weak character is exactly one that is currently holding back the unlock of the next character. The higher the upper threshold, the sooner a character is boosted after a mistake.

At most the **5** weakest are shown, worst first, with their error rate.

Exception, Listen with the Koch lesson: there a character counts as weak as soon as it has at least 8 attempts and its accuracy is **below the upper threshold** (exactly what holds back the unlock). Up to **10** are shown, in the “On the way to X” card (see below) when it is visible. Because the rate is tracked over time, a character stays weak until you get it right reliably again. This can take several blocks and sessions.

Tap a weak character to exclude it from the boost (it appears struck through), or to include it again.

- **Listen:** the weak characters already appear on the start view and on every result page. In the next block they are boosted at the *Moderate* level, with up to 3 draws per character (see Practice set). This only applies to random groups (Koch lesson or All characters · Random). With words, abbreviations, mixed and the practice set the boost has no effect, so the app does not show the weak characters there at all. If you have your own practice set characters and **Boost practice set** switched on, they are added, and the higher level applies (so Strong for all of them if you chose Strong).
- **Send:** the weak characters appear on the start view and on the result page when you practice **Random** (Koch lesson or all characters). With **Start** or **Next block**, the ticked characters come up twice as often in the following block. The boost lasts exactly one block; after that it is recomputed.

7.6.1 Practising recurring mix-ups

In addition to the weak characters, the app suggests mix-ups that keep recurring (e.g. K → R). The suggestions appear where the weak characters do (start view and result page in both) and only where the boost works (random groups). There are at most three pairs. A pair is suggested when

- the mix-up is **rising** or **new** (see the *Trends* tab of the character statistics), or
- it is simply **frequent**: at least 3 times in the last 14 days and in at least 10 % of the attempts at the played character. So a permanently bad pair is suggested before there is enough data for a trend.

Falling or vanished mix-ups are not suggested. **Tap a pair** to leave it out (struck through) or take it in again. Both characters of the pair then come up more often in the next block, with the same boost as the weak characters. The characters are **merged**: a character that is weak and in two pairs is boosted only once, the weights do not add up. Listen and Send count separately; in Listen only typing-mode blocks count.

7.6.2 Character selection in Send

In **Send · Koch lesson · Random** the characters are drawn exactly as in Listen (see Koch sequence): two draws in three come evenly from all active characters, every third from the last third. Weak characters you leave ticked in the **Weak characters** card are favoured in the next block; how strongly is set in the ⚙ sheet under **Boost practice set**. A mistake therefore does not act at once in the middle of a block but through the statistics in the next block, the same in Listen and Send.

7.7 Progress card “On the way to X”

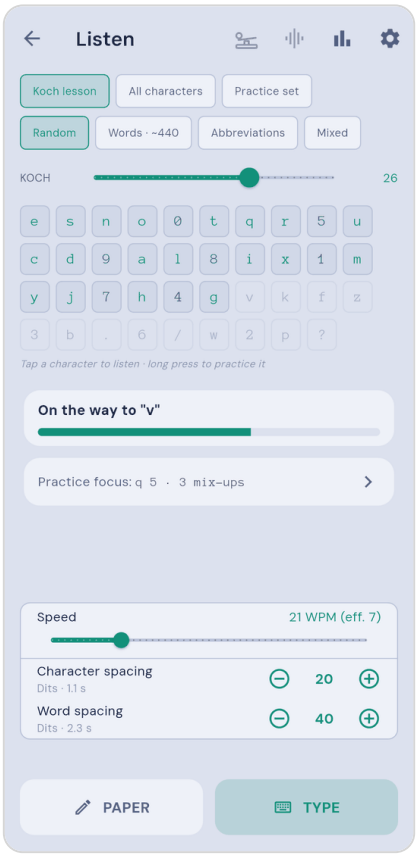
On the **Listen** result page with the Koch lesson, a card shows what is still missing before the next character:

- The **bar** shows how many of the required attempts you have made, added up over all characters that still lack attempts. If no attempts are missing any more and only accuracy is, it shows the weakest character’s accuracy relative to the threshold.
- **Still to practice (repetitions)** lists characters that still lack attempts, each as a small chip with the number missing, for example `q 7`. Those missing the most come first.
- **Accuracy below 90 % · practice more?** shows the weak characters with too low an accuracy, for example `y 84%`, the weakest first. These are the same characters that come up more often in the next block. **Tap a character** to exclude it (struck through). There is no separate “Weak characters” card in addition.

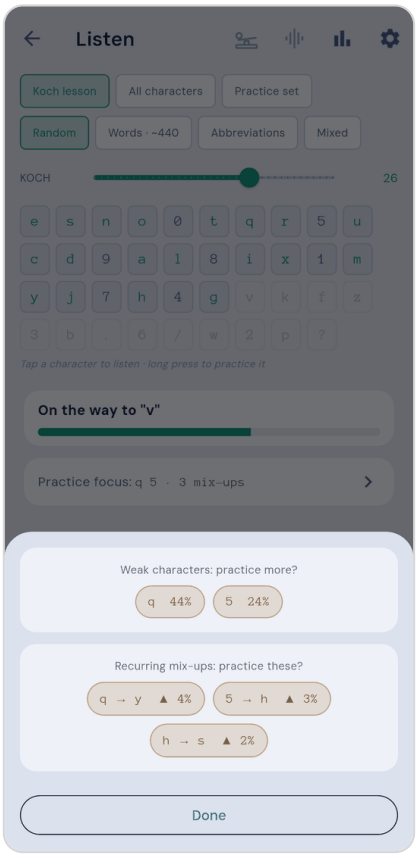
The “Still to practice” list shows at most 10 characters; the rest appear as “+N more”. If the card isn’t visible (shrunk, last Koch level, right after an unlock) or you aren’t practicing with the Koch lesson, the weak characters appear in their own card as before.

The **start page** of Listen (before the first block) shows the same card as at the end of a block, as long as everything fits on the screen. If space gets tight (small display, large font, many weak characters), it shrinks to just the title and the bar, without the character lists. The weak characters and mix-ups

then appear in **one row** (“Practice focus: E 5 Q · 2 mix-ups”). Tapping it opens a sheet from the bottom with all characters and pairs to tick or leave out; **Done** closes it. The same applies to the Send start view.



Listen start view with the compact “Practice focus” row



The row’s sheet: weak characters and mix-up pairs to tick or leave out

7.8 Trend

From the **sixth** block on, the status line shows a trend, for example “Trend 87 % ▲”:

- The number is the average block rate of the **last 5 blocks**.
- The arrow compares it with the average of the 5 blocks before. ▲ means at least 3 percentage points better, ▼ means at least 3 points worse, and ► means the same.

The trend is kept across sessions, separately for Listen and Send. The app stores the last 20 blocks for it.

7.9 Break hint

If errors pile up in **Listen** or **Send**, or your hit rate drops clearly during a session, the app suggests a short break on the result page. Fatigue and fading concentration show up this way early; a break often helps more than carrying on.

- It counts **characters**, not blocks – it makes no difference whether your block is 5 groups of 3 characters or 10 groups of 5. The check runs at the end of each block, with two rules:
  - **Cluster:** too many wrong characters within 20 consecutive ones (also across a block boundary). One missed group alone is never enough.
  - **Drift:** the hit rate of the latest characters is clearly below that of the first characters of the practice run (first against last 40, 60 or 80 characters). This only works once twice that many characters have been practised.
- You set the **sensitivity** in three steps:

Step	Cluster (wrong in 20 characters)	Drift (first against last … characters)
Early	from 7	40, drop of 12 percentage points
Normal	from 9	60, drop of 15 percentage points
Late	from 11	80, drop of 20 percentage points

  “Early” warns sooner but gives more false alarms; “Late” only warns on clear signs.
- Anything that makes the task harder is not counted as fatigue: a new character, higher speed, tighter spacing, other content, the interference (on/off and any change of its levels) or the input method restarts the observation. Only characters under the same conditions are compared.
- The hint appears **at most once per session**. A session ends after 10 minutes without a block. Listen and Send are watched separately.
- In **Send**, a word that was not right on the first try counts as wrong from its first wrong character on.
- It is never forced. **Break** returns to the start page, **Continue** closes the card, **Don’t show again** switches the hint off.
- Settings: **Settings → General → Break hint when the error rate rises** (on/off) and below it **Break hint: sensitivity**.

7.10 Adaptive mode settings

These are in the ⚙ sheet of **Listen** under **Adaptive mode**. They apply to Listen and Send together.

Setting	Meaning	Values
Success threshold low/high	Below *low* the pauses are lengthened. At *high* (2 blocks in a row) they are shortened or the speed is raised. *High* is also the accuracy each character needs for the Koch unlock	30–99 %, at least 5 points apart (**70 % / 90 %**)
EMA smoothing	How strongly the last block counts in the block EMA. Higher reacts faster, but is also jumpier	5–100 % (**30 %**)
Occurrences for unlock	Minimum attempts per character before the next Koch character can come	5–50 (**20**)

Some notes:

- The high threshold can't be set to 100 %. The moving error rate of a character that ever had an error never quite returns to 0. At 100 % that character would block the unlock forever.
- **EMA smoothing 100 %** means only the last block counts.
- If unlocking goes too fast for you, raise **Occurrences for unlock** or the high threshold. If it's too slow, lower them.

8 Daily goal and streak

Short, regular practice beats long, rare sessions. The app therefore counts your **active practice time** and shows it on the home screen.

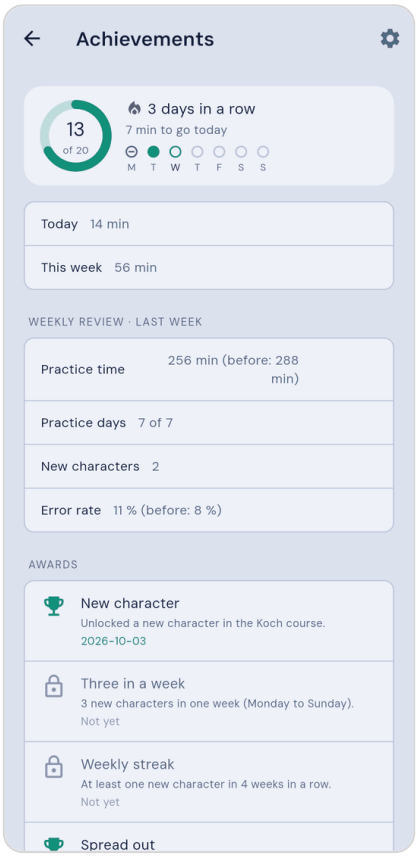
What counts. Time in the trainings (Listen, Send, games, adventure, QSO Bot, Own texts, Morse key) while you are active: you touched the screen or keyed within the last 20 seconds, or a block is playing (copying on paper). The CW Decoder, WiFi Trx and the reference screens (chart, tree) do not count. A day runs from 04:00 to 04:00, so practice after midnight still counts for the evening before.

The card. The ring fills up towards your daily goal and shows a check mark once it is reached. Next to it: the streak (days in a row on which you met the goal) and the week as dots: filled = goal met, ring = open or missed, a ring with a dash = free day. On a small screen or with a large system font the card shrinks to one line, so all tiles stay visible.

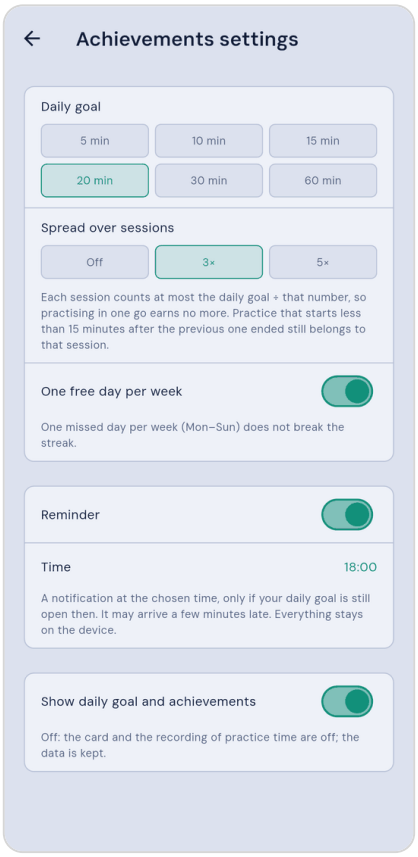
Free day. One missed day per week (Monday to Sunday) does not break the streak. It is not counted as a practice day, it only bridges the gap. A second miss in the same week ends the streak. You can switch this off.

Achievements page. Tap the card. It shows today and this week in minutes and below them the **awards** (see below). The gear icon opens its settings:

Setting	Meaning	Values
Daily goal	Minutes of active practice per day	5 / **10** / 15 / 20 / 30 / 60 min
One free day per week	Bridges one missed day per week	**On** / Off
Spread over sessions	Each session counts for at most the daily goal ÷ 3 or 5, so the goal is only reached when you practise spread over the day (see below)	**Off** / 3× / 5×
Reminder	A notification at the chosen time, only if your goal is still open then. The app asks for the notification permission when you switch it on	**Off** / On, time (**19:00**)
Show daily goal and achievements	Hides the card and stops recording. The data is kept; switch it on again here or under **Settings → General**	**On** / Off



Achievements page: daily goal, week and awards



Achievements settings

Weekly review. Below Today and This week a short card sums up one week: practice time, practice days, new characters and error rate, each with the week before in brackets. On Sunday it shows the running week, on every other day the last finished one (Monday to Sunday). The error rate appears from 3 blocks in the week. With no practice that week the card is not shown. No ranking, nothing leaves the device.

Awards. Ten of them, no pop-ups while you practise, only on this page. An earned award shows the day you first reached it; open ones are grey with a lock. Everything stays on the device.

Award	Condition
New character	Unlocked a new character in the Koch course
Three in a week	3 new characters in one week (Monday to Sunday)
Weekly streak	At least one new character in 4 weeks in a row
Spread out	3 sessions in one day (at least 5 min each, 15 min apart), also without the “Spread over sessions” setting
Spreading as a habit	Practised that way on 5 different days
Better than last week	Fewer errors than the week before, at least 3 blocks each
Clean run	3 blocks in a row with under 5 % errors

Award	Condition
Despite interference	A block with interference switched on and under 10 % errors
New speed	A new speed record in Listen or Send
Welcome back	Practised again after a break of at least 7 days

Tap an award: you see what it means, when you first and last reached it, and how often (open ones show only the explanation). What “how often” counts differs per award and is stated in the text there, e.g. new characters, weeks, days or blocks.

Error rates count Listen and Send together. Block awards only work with blocks played after this update.

Spread over sessions. Learning in small chunks works better than one long block. With 3× or 5× every session counts for at most the daily goal divided by that number (15 min and 3×: at most 5 min per session). Practising 45 minutes in one go therefore fills only a third of the ring, while 15 times one minute spread over the day fills it completely. Practice that starts less than 15 minutes after the previous one ended still belongs to that session. The ring shows the counted minutes. Once the current session has reached its share, the card says when practice counts again. Off by default, not everybody has time for several sessions a day.

Reminder. Optional and off by default. At the chosen time you get a friendly notification, but only if your goal for that day is still open. The app plans the next days in advance and re-plans whenever you finish a training or leave the app. Android may deliver it a few minutes late. It is scheduled on the phone itself, nothing is sent anywhere. If you refuse the notification permission, the reminder stays off.

9 Character statistics

The  **icon** in Listen and Send opens that training's statistics. Listen and Send have separate statistics.

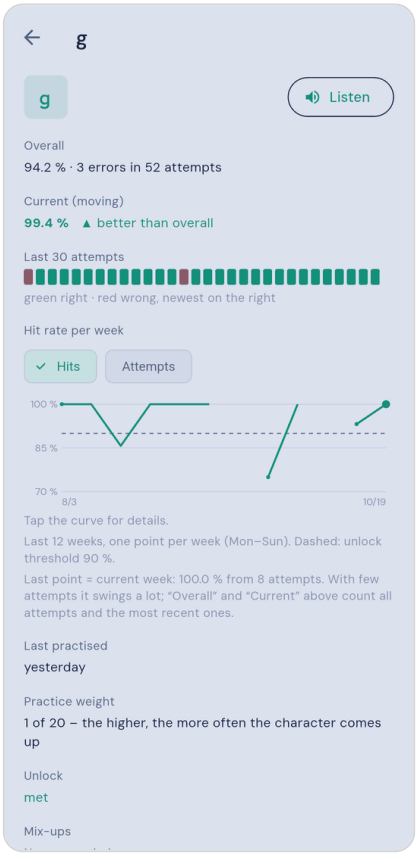
- **Listen** shows every active Koch character with its attempts (for example "14 attempts") and its **current** accuracy. A tick marks characters that meet the unlock condition, and an hourglass marks those that aren't there yet. Those also say what is missing ("6 more needed", "below 90 %"). The top shows "*x* of *y* characters ready". The characters that aren't ready come first, which helps when the unlock seems stuck.
- **Send** shows the same (attempts, current accuracy, tick or hourglass, "*x* of *y* ready"), but the characters are sorted by errors, the least reliable at the top. The unlock applies here too, for Koch with random characters. Mix-ups are not listed in the overview; you find them per character in the detail view.

The percentage is a moving average, not an overall rate. Newer attempts count more than older ones: each attempt goes into the value with 20 %. A miss therefore lowers it at once (100 % becomes 80 % at most), and correct answers rebuild it slowly. A character can drop below the threshold after a single miss even if it was flawless before. The tick needs both: enough attempts (default 20) **and** a current accuracy above the threshold (default 90 %). The bar shows only the attempts, not the accuracy.

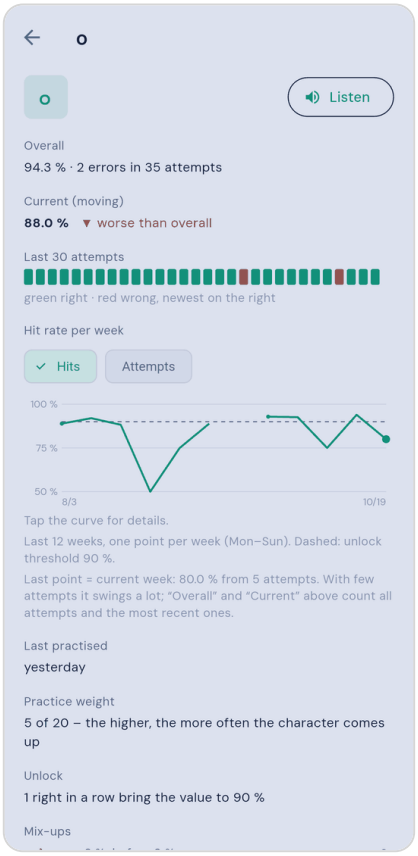
Tap a character (a hint on the list says so) to open its detail view as a page of its own, with a back arrow at the top left. It shows:

- **Overall** – accuracy over all attempts, with errors and attempts.
- **Current (moving)** – the value from the list, with an arrow: ▲ better, ▼ worse or ► same as the overall rate (from 3 percentage points apart, and only from 5 attempts). It shows whether a dip was a slip.
- **Last 30 attempts** – a strip of green (right) and red (wrong) bars, newest on the right. It fills up as you practise; older data has no history.
- **Hit rate per week** – a curve of the last 12 weeks for this character (tab *Hits*; dashed: the unlock threshold) or the number of attempts per week (tab *Attempts*). Each point is one calendar week (Mon–Sun), the last one the current week; below it you see its rate and number of attempts. Tap a point (tab *Hits*) or a bar (tab *Attempts*) to see that week's attempts and hit rate below the chart. The last point can therefore differ clearly from *Overall* and *Current*, especially with few attempts. Weeks without practice are left out.
- **Last practised** – today, yesterday or *n* days ago (also only from the next practice on).
- **Practice weight** – from 1 to 20. The higher, the more often the character comes up in adaptive exercises.
- **Unlock** – what is still missing: how many attempts and how many right answers in a row until the rate is back above the threshold.

- **Mix-ups** – what you sent (Send) or typed (Listen) instead. Only the first attempt of each word counts, and in it the first wrong character. In Listen this exists only in typing mode; in paper mode only “wrong” is marked, not what you heard. A skipped word (pass) does not count. Mix-ups are also logged per day, separately for Listen and Send. An arrow after each pair shows its trend (see the *Trends* tab), and below it a curve of the mix-up rate over 12 weeks; pick other pairs with the chips.
- **Listen** – plays the character three times, at this training’s speed.



Detail view in Listen: weekly curve, unlock



Detail view in Send: last 30 attempts, curve, mix-ups

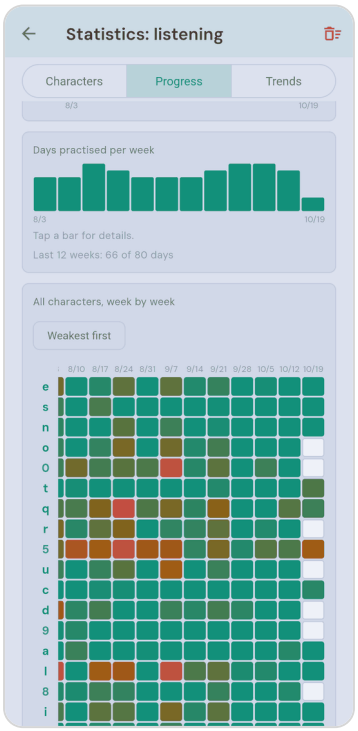
Progress tab. At the top, switch between *Characters* (the list above), *Progress* and *Trends* (see *Trends* below). Progress shows how practice develops over time, separately for each training. It fills up from the day this feature was added; earlier practice is not included. Choose **4 weeks**, **12 weeks** or **All**:

- **Practice time** right at the top for the chosen period, as “3 h 25 min”; with *All* from 24 hours on also with days (“2 d 3 h 25 min”). Two tiles: the time of **this training** (Listen or Send) and the **total of all trainings**. It is the same active practice time the daily goal counts; it runs while you practise actively and doesn’t depend on breaks or gaps. The time per training has only been recorded separately since this update; for earlier days it is only in the total (a note gives their number). Before the daily goal there is no time at all.
- **Hit rate, WPM and days practised** below it. Hit rate and WPM are those of the latest day (4 weeks), week (12 weeks) or month (All, from about half a year on) with practice, weighted by attempts, with an arrow against the period before. “Days practised” reads like *42/88*.

- **Hit rate per day / week / month** and **speed (WPM)** as curves. Periods without practice are gaps, not zero. Tap a point of the hit rate to see that period's attempts and rate.
- **Days practised**: one box per day (4 weeks), otherwise a bar per week or month. Tap one to see the number of days, this training's time and the total time of that period.
- **All characters, week by week**: a heatmap, characters down, weeks across, the characters coloured by kind (letters, digits, punctuation/prosigns) so O and 0 are easy to tell apart, from red (below 60 %) to green (from 90 %). A cell with fewer than 5 attempts stays empty. The columns fill the whole width; if the weeks don't fit (All), the cells scroll and start at the newest week; the characters and week headings stay in place. **Weakest first** sorts by the latest hit rate. Tap a character to open its detail view.



Progress tab (Listen): hit rate, speed and days practised over 12 weeks



Below it the heatmap: all characters, week by week



Progress tab (Send)

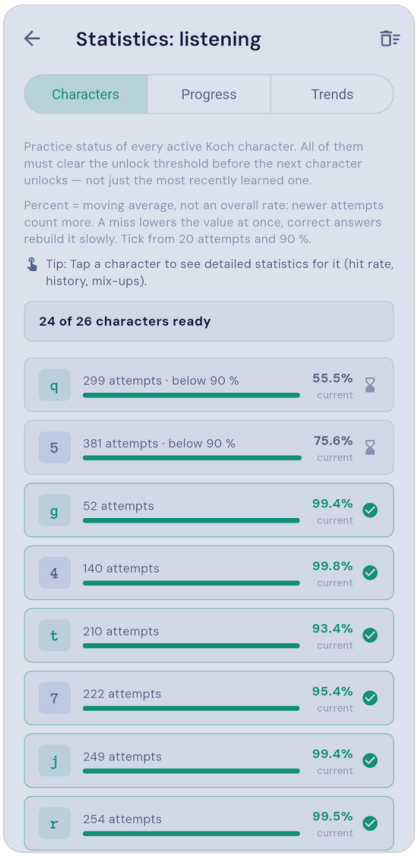
Trends tab. Shows whether a specific mix-up is becoming more or less frequent, separately for Listen and Send. It compares the **mix-up rate** of the last 14 days (or 30 days, chip) with the days before. The rate is "how often K was taken for R, divided by all attempts at K", so practising more does not make a mix-up look more frequent. The pairs are grouped by what needs attention:

- **Getting more frequent** (▲) and **Newly appeared** – worth practising.
- **Getting less frequent** (▼) and **Gone** (✓) – practice is working.
- **Unchanged** (►) – the rate stayed about the same (change under 2 percentage points or under 30 % of the old rate).
- **Too little data** – collapsed. An arrow only appears once the character came up at least 15 times in both periods and there are at least three mix-ups in total.

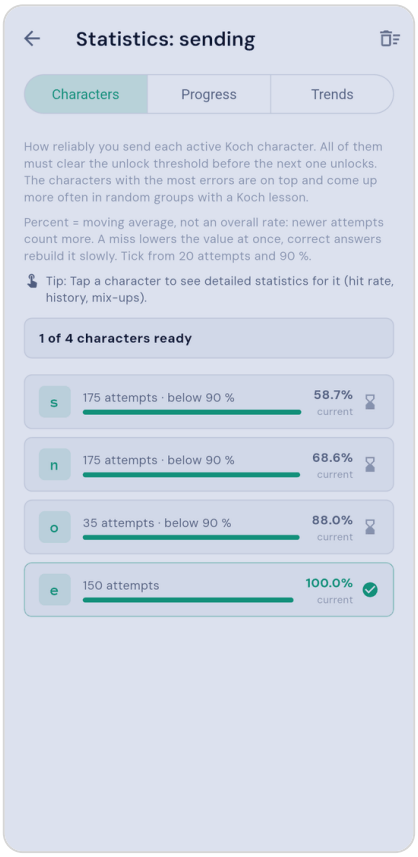
Each row shows the pair, arrow, current rate, the rate before, a small weekly curve and the count in the current period. Tap it to open the character. Mix-ups are logged per day only since this feature

exists, so the first weeks show just a “Trend from …” hint with a progress bar; there is no trend for the time before.

The **reset** icon deletes all of this training’s statistics after a confirmation: error rates, weights, mix-ups and the progress. The other training’s statistics are kept. This cannot be undone.



Listen statistics: attempts, accuracy, ready



Send statistics with common mix-ups



“Trends” tab (Listen): new, growing and fading mix-ups



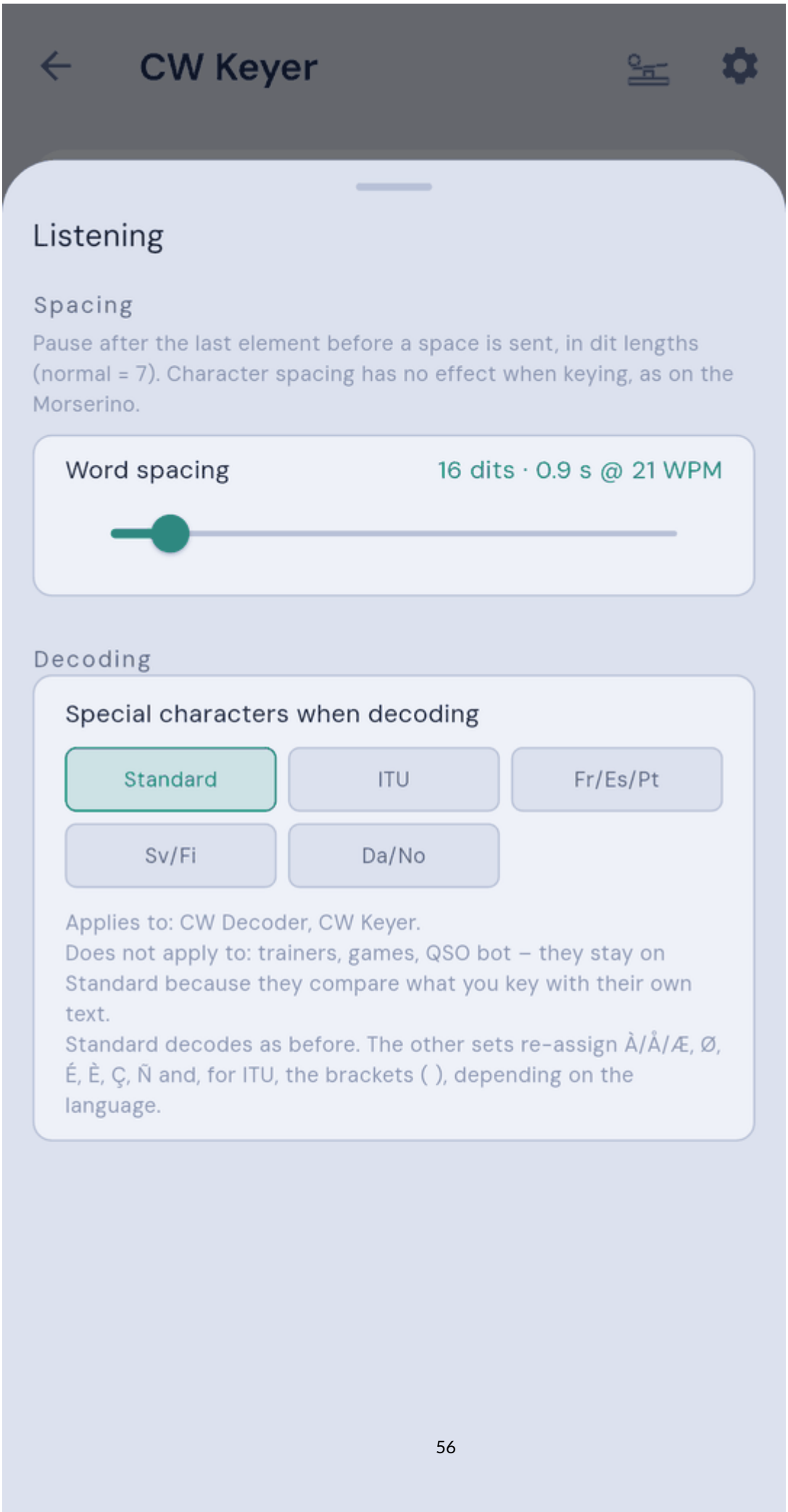
“Trends” tab (Send)

10 CW Keyer

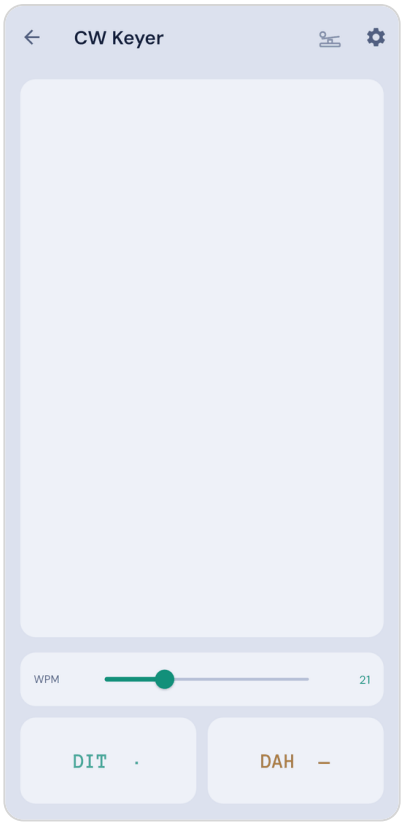
For free keying. What you key is heard and shown as decoded text.

- Use the **keys** at the bottom (DIT on the left, DAH on the right), or a real Morse key through an adapter (see Morse key). In **Straight** mode a single **KEY** area appears instead.
- **WPM** sets the keyer speed, 5 to 60 WPM.
- The text runs from the bottom upwards. You can scroll back to older lines, and you change the text size with two fingers.
- The ⚙ sheet at the top right holds the **word spacing** (word spacing, default 7 dits). It sets the pause after which a space is inserted. Character spacing has no effect when keying, as on the Morserino.
- **What appears in the text:** a pattern that is no character shows as * in the text, as on the Morserino. A real question mark (`..--..`) stays **?**. Seven or more dits in a row are the error sign and show as **ERR**. Besides letters, digits and prosigns the keyer also recognises Ä, Ö, Ü, CH and `;` `!` `"` `'`.
- The ⚙ sheet also holds **Special characters when decoding** (see Special characters when decoding).

::: {.shots one}


CW Keyer
Listening
Spacing
Pause after the last element before a space is sent, in dit lengths (normal = 7). Character spacing has no effect when keying, as on the Morserino.
Word spacing
16 dits · 0.9 s @ 21 WPM
Decoding
Special characters when decoding
Standard
ITU
Fr/Es/Pt
Sv/Fi
Da/No
Applies to: CW Decoder, CW Keyer.
Does not apply to: trainers, games, QSO bot – they stay on Standard because they compare what you key with their own text.
Standard decodes as before. The other sets re-assign À/Å/Æ, Ø, É, È, Ç, Ñ and, for ITU, the brackets ( ), depending on the language.

You set the keyer mode and its details in the global settings under **Keyer** (see Keyer). They apply everywhere you key: CW Keyer, Send, WiFi Trx, QSO Bot and the games.



CW Keyer with decoded text

11 CW Decoder

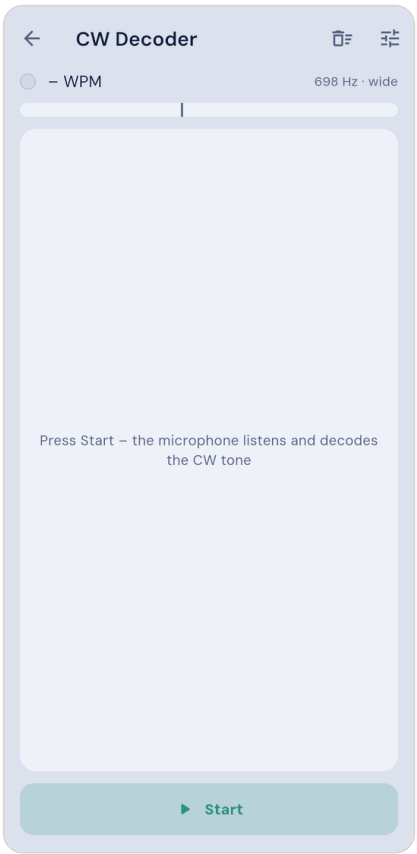
The CW Decoder listens through the **microphone** and writes down what it recognises as Morse code. The signal might come from a radio, a practice program or another Morserino.

On first start Android asks for microphone permission. Without it the decoder can't work. If you declined, allow the microphone in the Android settings under **Apps → APEX CW → Permissions**.

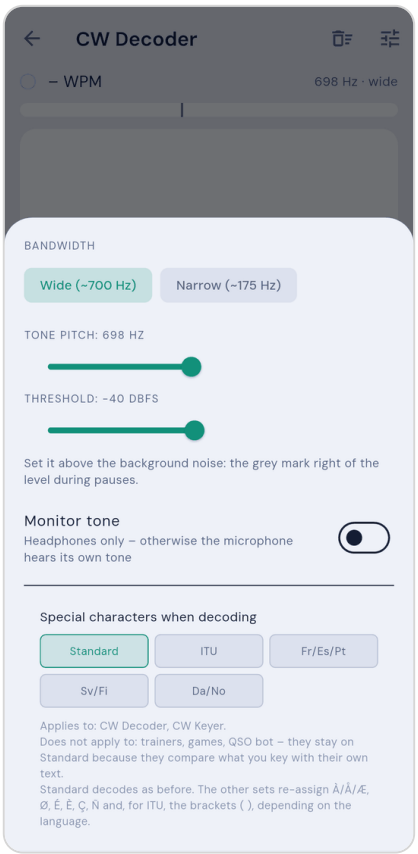
11.1 Operation

- **Start / Stop** switches copying on and off.
- The **status line** shows the set pitch, the bandwidth and the detected speed in WPM.
- The **level meter** shows the loudness of the tone at the set pitch. A grey mark shows the noise level during pauses.
- The icons at the top right clear the text and, with the sliders icon, open the decoder settings.

The decoder adapts to the speed automatically, so there's nothing to set. It comes from the firmware. A Goertzel filter detects the tone, and timing logic tells dits, dahs and pauses apart. It keeps learning the current dit and dah length as it goes.



CW Decoder listening, with level meter



Decoder settings

11.2 Decoder settings

Setting	Meaning	Values
Bandwidth	**Wide** is more tolerant of a pitch that doesn't quite match. **Narrow** filters interference better, but the pitch must be exact	**Wide (~700 Hz)** / Narrow (~175 Hz)
Tone pitch	The frequency of the CW tone the decoder listens for	300–1200 Hz (**698 Hz**)
Threshold	Minimum loudness for a tone to count. Set it just **above** the grey noise mark	−65 to −10 dBFS (**−40 dBFS**)
Monitor tone	Plays back the detected tone cleanly at the set pitch	**Off** / On
Special characters when decoding	How codes that mean different things in different languages are decoded, see below	**Standard** / ITU / Fr/Es/Pt / Sv/Fi / Da/No

Tips

- If you don't know the signal's pitch, start with **Wide** and turn the pitch until the level jumps clearly on the characters. Then switch to **Narrow**.
- Random `e` and `t` appearing in the pauses mean the **threshold** is too low.
- Only use the **monitor tone** with headphones. Otherwise the microphone hears the tone itself and confuses the decoder.

11.3 Special characters when decoding

Some Morse codes mean different letters in different languages, and the brackets exist only in the ITU recommendation (ITU-R M.1677-1). This setting (in the ⚙ sheet of the **CW Decoder** and the **CW Keyer**) chooses how they are decoded. It is shared by both and stays saved.

Applies to: CW Decoder and CW Keyer. **Does not apply to:** Send, Listen, the games and the QSO Bot. They always decode with **Standard**, because they compare what you key with their own text (e.g. `<KN>` or AE).

Code	Standard	ITU	Fr/Es/Pt	Sv/Fi	Da/No
`.--.-`	*	*	À	Å	Å
`.-.-`	Ä	Ä	Ä	Ä	Æ
`---.`	Ö	Ö	Ö	Ö	Ø
`-.--.`	KN	(	KN	KN	KN
`-.--.-`	*	)	*	*	*

Code	Standard	ITU	Fr/Es/Pt	Sv/Fi	Da/No
..–..	*	É	É	*	*
.–..–	*	*	È	*	*
–.–..	*	*	Ç	*	*
––.––	*	*	Ñ	*	*

Standard decodes as before. All other codes, Ü (..––) included, are the same in every set. These letters cannot be sent yet; the setting only affects decoding.

12 WiFi Trx

With WiFi Trx you send Morse over the internet or your local network with other Morserinos, apps and servers. The app uses the same protocol as the Morserino ("Morse over Packet", MOPP, UDP port 7373). One example is the chat server **cq.morserino.info**.

The app uses the phone's existing WiFi or mobile data connection, so there are no WiFi credentials to enter.

12.1 Services

At the top you choose the **service** to connect to. The default is `cq.morserino.info`.

- **+** adds a new service, and the **pencil** edits or deletes the selected one. Each service has a **Name** and a **Server** (host name or IP address).
- An **empty server** means broadcast to the local network. This lets you exchange Morse with Morserinos on the same WiFi.
- You can only do this while not connected.

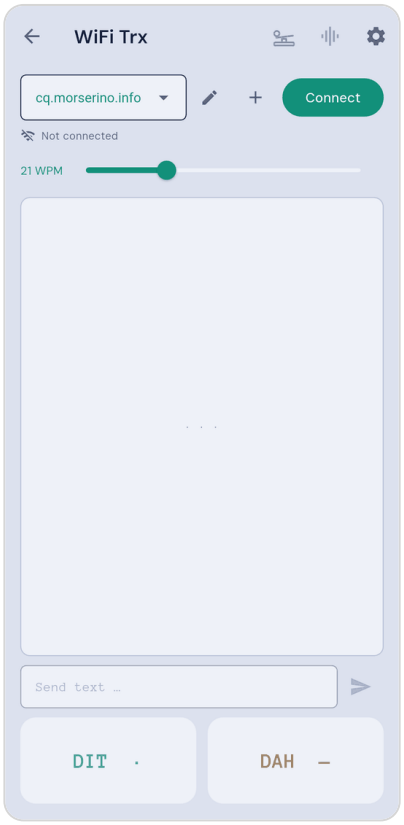
Connect connects and **Disconnect** disconnects. The line below shows the status. If the server ends the connection (`:bye`), the app disconnects too.

12.2 Sending and receiving

- **Send** with the touch keyer or the adapter. Each word is sent as a packet after the word pause. You can also type text into **Send text…** and send it. It goes out at the set speed.
- **WPM** sets your sending speed.
- **Receiving:** received words are played one after another at the **sender's** speed and shown in the log.
- The **log** tells received (RX) from sent (TX) text and is kept per service. **Long-press the log** to clear it after a confirmation.
- In the ⚙ sheet you set the **word spacing**, as in the CW Keyer. It decides after which pause your word is sent.

WiFi Trx only works while the app is in the foreground.

If your partner is behind a different router, UDP packets on port 7373 must be able to get through. With a public server such as cq.morserino.info this is normally no problem.



WiFi Trx with service, log and text field

13 QSO Bot

The QSO Bot is a simulated partner for complete, realistic radio contacts, without going on the air. The bot sends in CW, understands what you key and reacts to it. It only sounds through the app's speaker and never sends anything to the network.

13.1 Starting

1. Choose the **QSO type**: **SOTA/POTA**, **Standard** or **Contest**.
2. Press **Start**. The bot listens for 5 seconds:
   - **If you call CQ** (for example `cq cq de oe1abc k`), it answers you.
   - **If you stay silent**, it calls CQ itself and you answer.

Each contact uses a new, realistic call sign for the bot.

13.2 QSO types

- **SOTA/POTA** is a summit or park activation with report and reference. If the bot calls CQ, it is the activator and you are the chaser. If you call CQ, it's the other way round. A session is a single contact.
- **Standard** is the classic QSO in three rounds. First come report, name and QTH. Then station details follow (rig, ant, wx, age), and finally the sign-off. The bot finds your details by the keywords `name`, `qth`, `rig`, `ant`, `wx` and `age`, and it remembers your name.
- **Contest** is many very short contacts in a row. The exchange depends on the **Contest type**: **CQ WW** (report + CQ zone) or **WPX** (report + serial number). The session ends by itself after a longer silence.



A QSO with the bot, answered with the text field

13.3 Talking to the bot

- **End each over** with `k`, `bk`, `<ar>`, `<sk>` or `73`, or simply pause. The bot won't talk over you.
- **Repeat:** `agn`, `rpt` or `?` repeats the bot's last over. `rpt rst`, `rpt call`, `rpt qth` or `rpt name` repeats just that item.
- **Bot speed:** `qrs` makes it slower and `qrq` faster. Your own speed doesn't change.
- **Correcting:** `<err>` (eight dits) or `eeee` discards your last word.
- You can send reports in cut numbers: `5nn` for 599, `t` for 0, `a` for 1, `n` for 9.
- Filler words like `de`, `r` and `ur` don't matter, because the bot picks out the information.

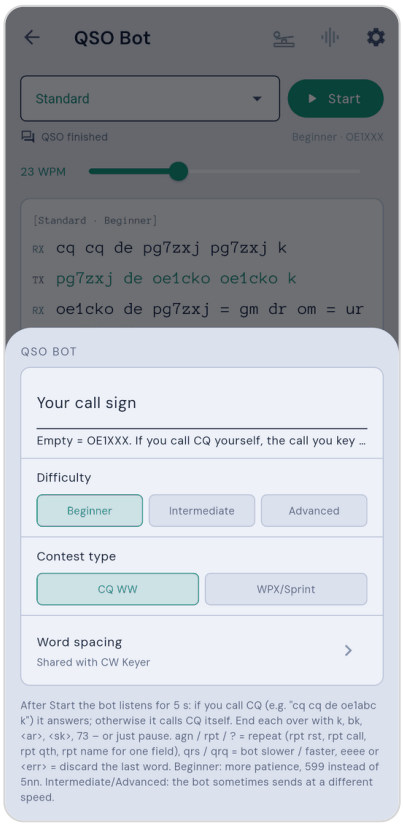
- Instead of keying, you can type text into the input field. It is treated as if keyed, but not played aloud.

13.4 QSO Bot settings

These are behind the ⚙ icon in the QSO Bot:

Setting	Meaning	Values
Your call sign	Your call sign for the bot. If you call CQ yourself, the call you key is used	empty = OE1XXX
Difficulty	**Beginner**: more patience, 599 instead of 5nn, and the bot sends at your speed. **Intermediate** and **Advanced**: the bot sometimes calls at a slightly different speed, as practice for qrs/qrq. Advanced also keeps a tighter rhythm	Beginner / **Intermediate** / Advanced
Contest type	Exchange in a contest	**CQ WW** / WPX
Word spacing	After which pause your word counts as finished. It is shared with the CW Keyer	6–105 dits (**7**)

WPM sets your speed. **Long-press the log** to clear it after a confirmation.



QSO Bot settings

14 Own texts

Own texts lets you listen to any text as Morse code – a weather report, a newspaper article, a book chapter or a list of call signs. You find it under **Free modes**.

14.1 Adding texts

1. In any app, copy a text to the clipboard.
2. In **Own texts**, tap **Paste from clipboard**.
3. Give the text a title (the first words are suggested) and tap **Add**.

Even quicker: select the text in any app (browser, mail, notes, e-reader), tap **Share** and choose **APEX CW**. The app opens **Own texts** with the same dialog for the title. The same size limit and character handling apply as for pasting.

You can add as many texts as you like. A text can be at most 20,000 characters long. The ⋮ menu next to a text renames or deletes it (after asking). You can't edit a text in the app: to change one, paste the corrected version. The texts stay on your device.

14.2 What is sent

The text is always shown as you pasted it. Only the audio replaces characters Morse doesn't have:

- Umlauts and ß: Ä → AE, Ö → OE, Ü → UE, ß → SS. Other accents are dropped (é → E).
- `!` becomes `.`, `;` becomes `,`, `&` becomes AND, dashes become `–`. A dash standing alone between spaces is skipped.
- Quotation marks, brackets and other special characters are not sent.
- Write **prosigns** in angle or square brackets, e.g. `<KA>` or `[SK]`; they are sent as one character. A pair of letters without brackets is always two letters.

A paragraph break in the text is sent as a double word gap.

14.3 Playing

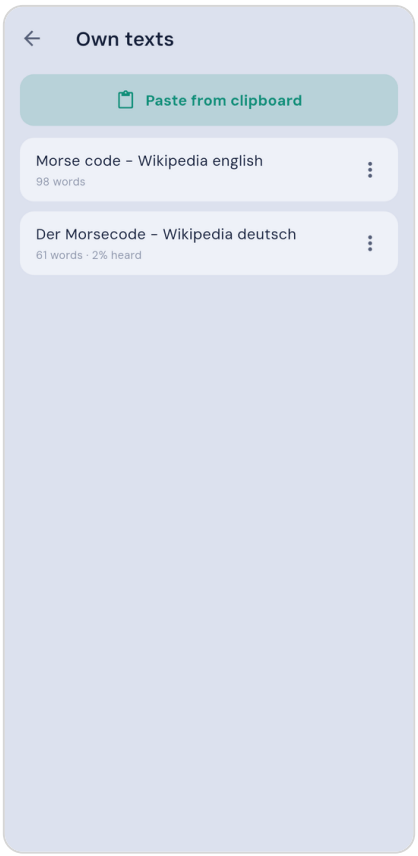
Tap a text, then ▸. At the top you set the **speed** (WPM) with the slider or ⊖/⊕ – also **while** the text plays; the change applies at once.

Button	Effect
▸ / ▮▮	Play; pause (then it goes on from the start of the current word)
Word	Repeat the current word, then pause
Sentence	Repeat the current sentence, then pause

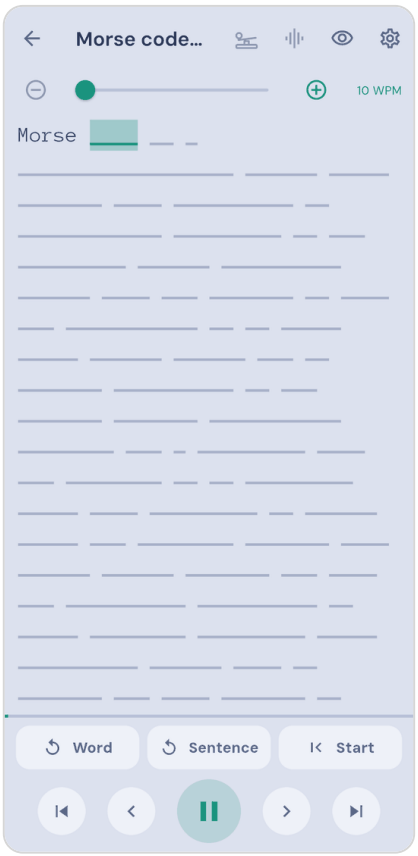
Button	Effect
Start	The whole text from the beginning
⏮ / ⏭	To the start of the previous / next sentence
‹ / ›	One word back / forward

Tap any word in the text to listen from there. Pinch with two fingers to make the text larger or smaller. At the end the text starts again from the beginning.

Progress is saved automatically. When you open a text you have started, the app asks: **Continue** or **Start over**. The list shows how much of each text you have heard.



Own texts: the library



A text playing (display: After playing)

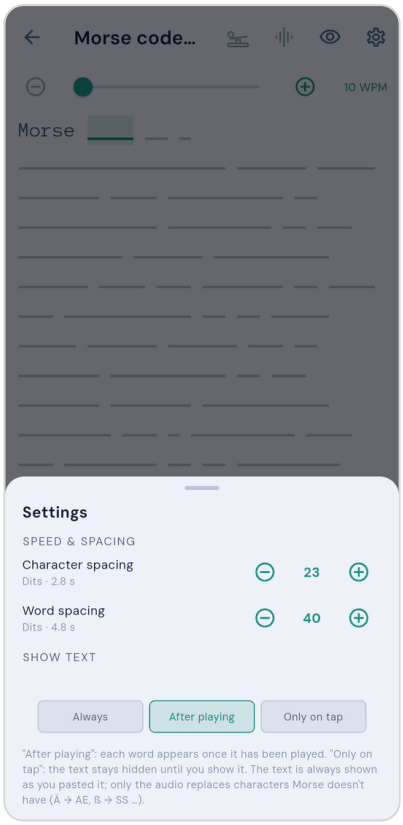
14.4 Settings (⚙)

- **Character spacing** and **word spacing** in dits, as in the trainings. They also apply while playing.
- **Show text**: **Always**; **After playing** (each word appears once it has been sent – the default); **Only on tap** (the text stays hidden until you show it with the eye icon at the top).

Speed, spacing and display apply to all texts. Pitch and tone softness come from the general settings. The display stays on while you are in a text.

Keep listening in the background: A text that is playing goes on when you turn the screen off or switch to another app. While it plays, Android shows a notification with **Previous sentence**, **Play/**

Pause and **Next sentence**; the same buttons appear on the lock screen and on Bluetooth headphones (previous/next jump by sentence). The first time, Android asks for permission to show notifications – without it the text still plays on, but the notification has no buttons. A phone call, another audio app or unplugging the headphones pauses the text. Leaving the text with “Back” ends playback. This applies to Own texts only, not to the other exercises.



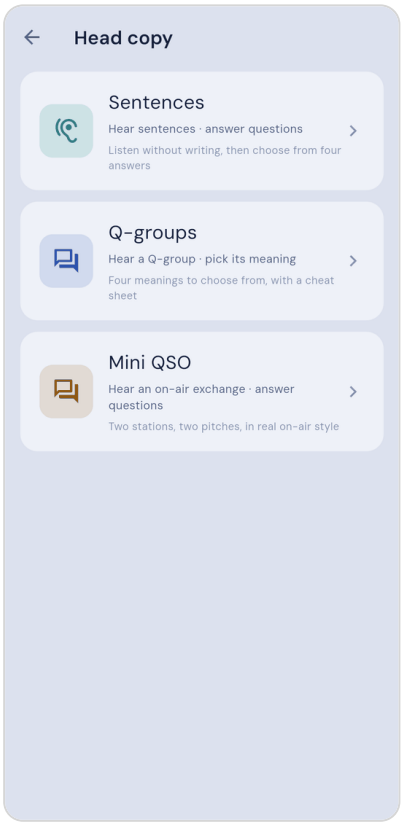
⚙ sheet of Own texts

15 Games

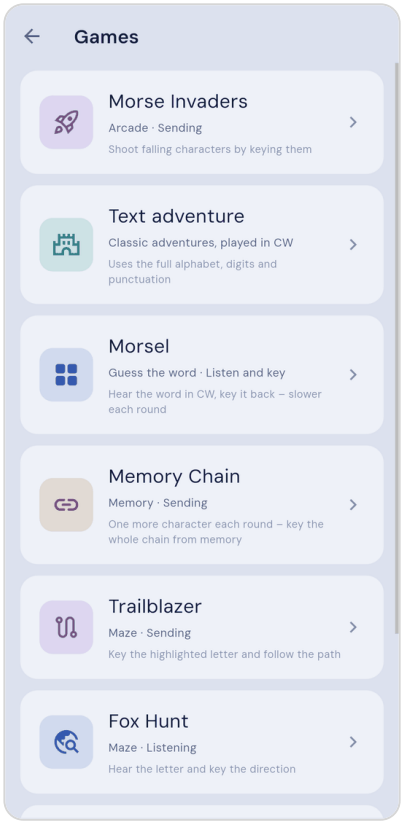
Under **Play → Games** there are eight entries. The list is longer than the screen; a scroll bar at the edge shows that it goes on. You play Morsel, Morse Invaders, Memory Chain, Trailblazer, Fox Hunt, Fight the Pileup and Radio Cave with the touch keyer or the adapter. They use the keyer settings. Morsel, Morse Invaders, Memory Chain, Trailblazer and Fox Hunt take the Koch lesson from **Send**, and Morsel, Memory Chain, Trailblazer and Fox Hunt also let you change it for the game only. Each of these seven shows short rules before you start. The text adventure has its own speed and input settings; it also takes the keyer mode from the keyer settings.

15.1 Head copy

Under **Practice → Head copy** you pick one of three listening modes: **Sentences**, Q-groups or Mini QSO. This section describes **Sentences**.



Head copy: the three listening modes



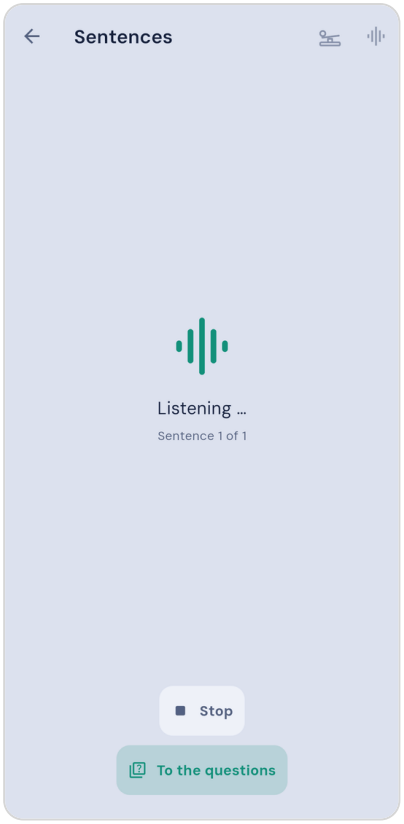
The games

Here you practise **understanding** Morse without writing it down. You hear one to three short sentences, then answer questions with four answers each. The text stays hidden until the result.

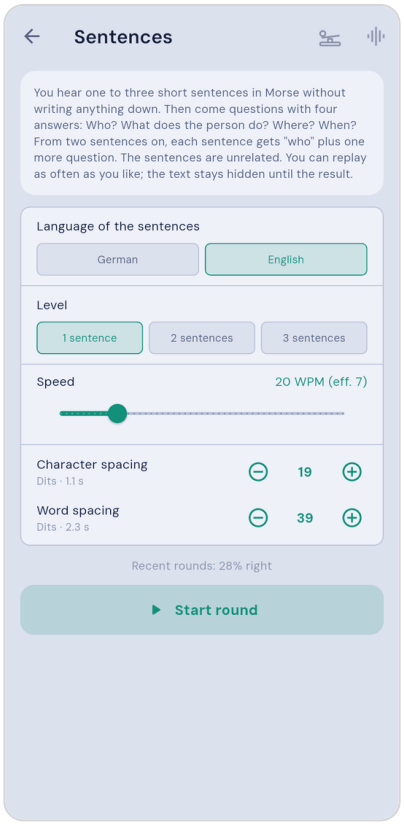
Start. Choose the **language of the sentences** (German or English; it is independent of the app language), the **level** (1, 2 or 3 sentences) and the **speed**, plus **character spacing** and **word spacing** (in dits, with the time in seconds below). They apply to Sentences only; on first open the app takes them from **Listen**, after that they are separate. The **effective speed** is shown in brackets behind the speed, the result of speed and both spacings (formula under Effective speed). Below the start button you see the hit rate of the recent rounds.

Listening. The sentences play one after the other with a short pause. **Listen again** replays the whole round as often as you like. **To the questions** goes on. The interference simulation (icon at the top) works as in the other trainings.

Questions. For each sentence you are asked for its parts, in the order **Who? – What does the person do? – Where? – When?** (When only if the sentence has a time). A follow-up question names the correct answers of the earlier questions, even if you got one wrong, and never anything asked later. After you tap you see at once what was right. **Level 1** asks every part. **Levels 2 and 3** ask “Who?” plus one more question per sentence, prefixed with “Sentence N:”. The sentences are unrelated. You can listen to the sentences again during the questions too.

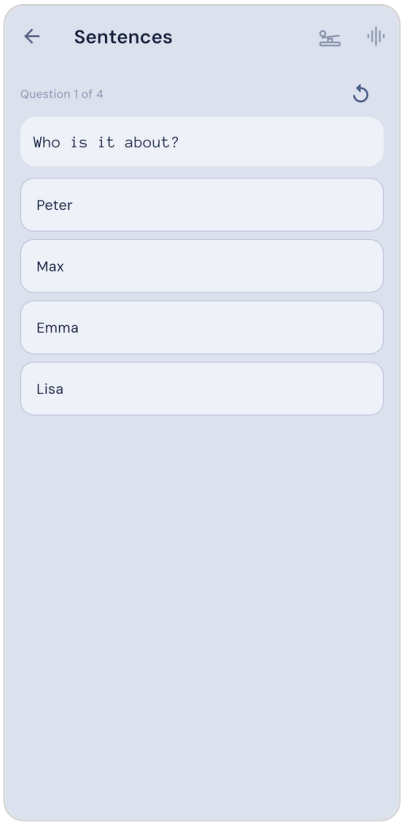


Sentences: the round plays, the text stays hidden



Sentences: start view with language, level, speed and spacing

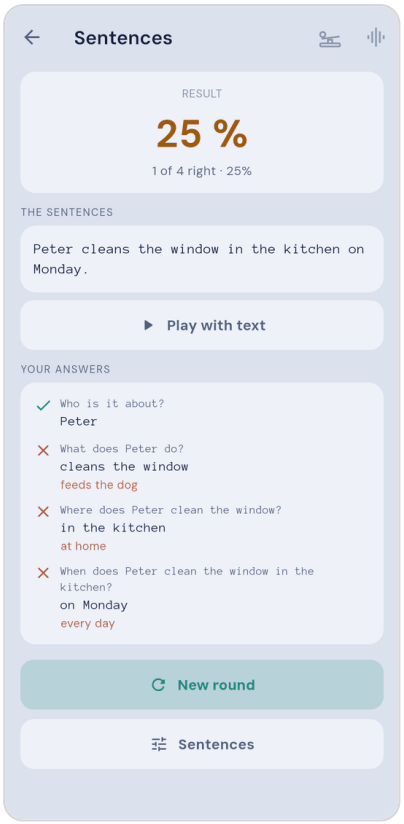
Result. The round's hit rate (right answers out of all questions), the sentences in plain text, **Play with text** (the sentence playing is highlighted) and your answers. The sentences are built from blocks and always make sense: places fit the activity. German umlauts are sent as AE, OE, UE, so "Küche" goes out as KUECHE. The time counts towards the daily goal.



Sentences: a question with four answers

15.2 Q-groups

Here you learn to **understand Q-groups** (QTH, QRZ, QRM …) at once when you hear them. You hear a group in Morse and pick from four answers what it means. Like Sentences, it needs no key.

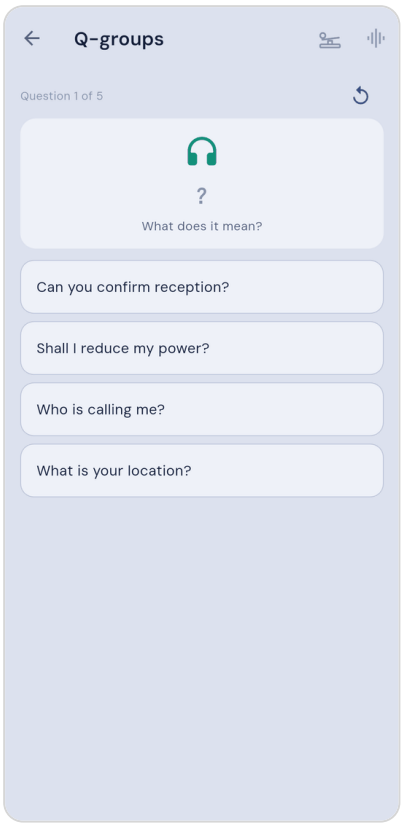


Sentences: result with the plain text and your answers

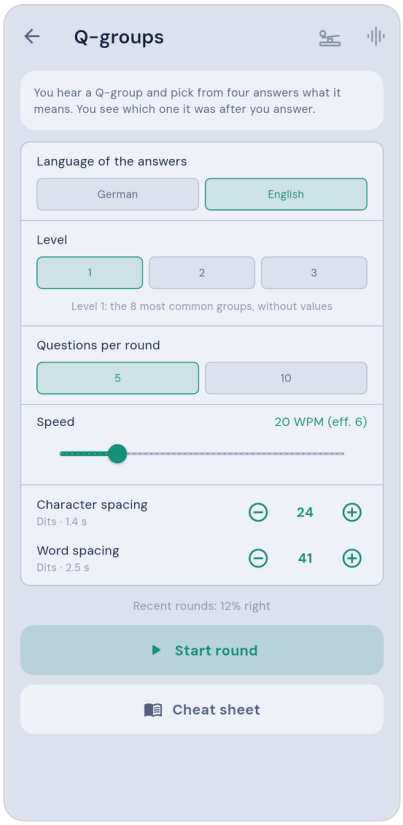
Start. Choose the **language of the answers** (German or English, independent of the app language), the **level** (1, 2 or 3; the groups are ordered by how common they are, and each level adds the rarer ones: level 1 has the 8 most common, level 2 has 16 in all, level 3 all 24; below the choice it says what the level means), the **questions per round** (5 or 10), **speed**, **character spacing** and **word spacing**. They apply to Q-groups only; on first open the app takes them from **Listen**. The **effective speed** is shown in brackets behind the speed, below the start button the hit rate of the recent rounds. **Cheat sheet** opens the list of all groups.

Questions. The group plays by itself when the question appears; the button at the top right plays it again. You only see which group it was after you answer. Some groups also come as a **question** (“QRZ?”); then the four answers are questions too. QRM, QRN and QSB only exist as statements. From level 2 on, some are followed by a value, like “QTH WIEN” or “QRG 7030”. The answers come from the groups of your level; two similar meanings (for example QRM and QRN, QRS and QRQ) never appear together in one question.

Result. The round’s hit rate and every question with the group played, the right meaning and your answer. Tap a row to hear the group again. The time counts towards the daily goal.

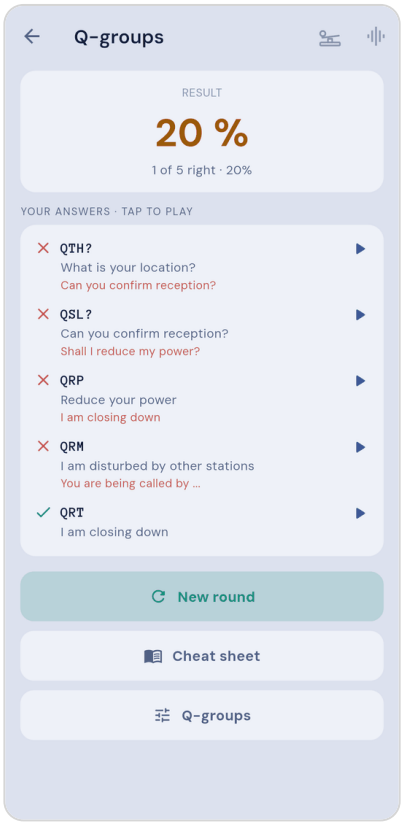


Q-groups: the group has played, four meanings to choose from



Q-groups: start view with level, questions per round and speed

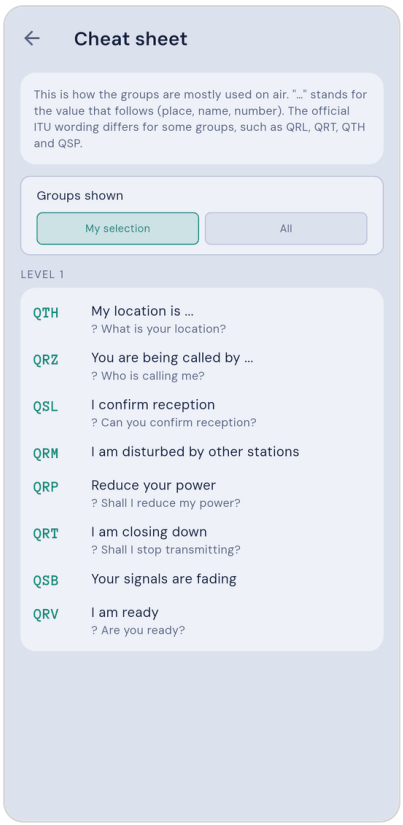
Cheat sheet. All groups with their meaning, ordered by level, with the question form below (marked with “?”). You reach it from the start and the result, not during a question; **All** also shows groups outside your level. The meanings are worded the way the groups are used on air. The official ITU wording differs for some: QRL there means “Are you busy?”, QRT “Stop sending”, QTH gives latitude and longitude, and QSP asks for relaying free of charge. For QRP, QRO, QRS, QRQ and QSY the statement is a request to the other station (“Send slower”).



Q-groups: result with the group played and its meaning

15.3 Mini QSO

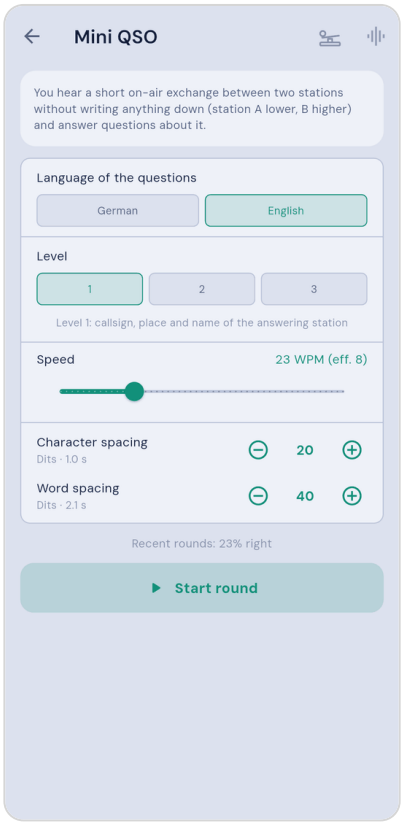
Here you hear a short **on-air exchange** between two stations in real radio style (“CQ CQ DE OE1ABC K” … “UR 579 QTH BONN NAME TOM BK”) and answer questions about it. As with Sentences you write nothing down and need no key. The abbreviations are the same in both languages; only the questions are translated.



Q-groups: the cheat sheet

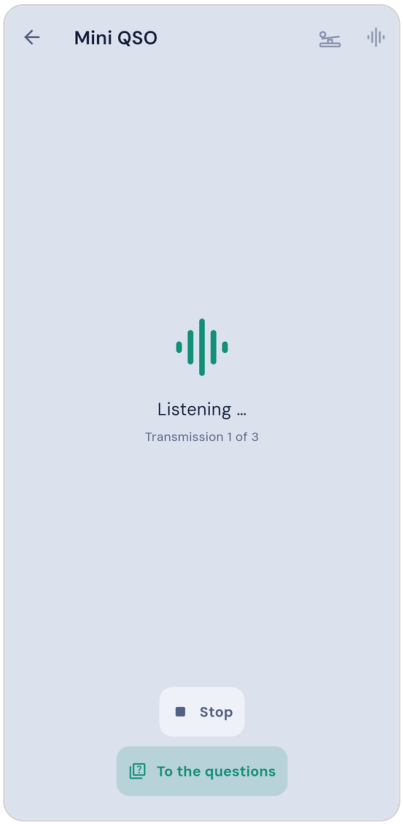
Start. Choose the **language of the questions** (German or English, independent of the app language), the **level** (1, 2 or 3; below the choice it says what the level means), **speed**, **character spacing** and **word spacing**. They apply to Mini QSO only; on first open the app takes them from **Listen**. The **effective speed** is shown in brackets behind the speed, below the start button the hit rate of the recent rounds.

Listening. Station A (it calls CQ) sounds a little lower, station B (it answers) a little higher than your set pitch, like two signals in the band. The callsigns are random ones as in the QSO bot; names and places are made up. You can hear the exchange as often as you like; the text stays hidden. **To the questions** goes on.



Mini QSO: start view with level and speed

Levels.

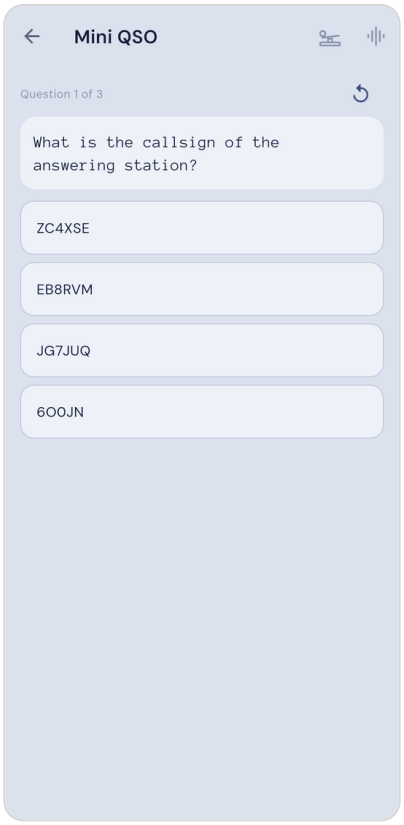


Mini QSO: the exchange plays, the text stays hidden

Level	Exchange	Questions
1	call and answer	callsign, place (QTH) and name of the answering station
2	as level 1, plus report and one piece of station info (RIG, PWR, ANT or WX)	also report and station info (5 questions)
3	full exchange of both sides with closing	both callsigns and four randomly chosen items from both sides (6 questions)

Questions. Four answers to choose from, always in the order you heard the facts. In levels 1 and 2 the station is called “the answering station”. In level 3 the first two questions ask for the callsigns; after that every question names the station by its callsign (“What is the name of DL2XYZ?”), even after a wrong answer. So no question gives away something that comes later, and one mistake does not spoil the rest.

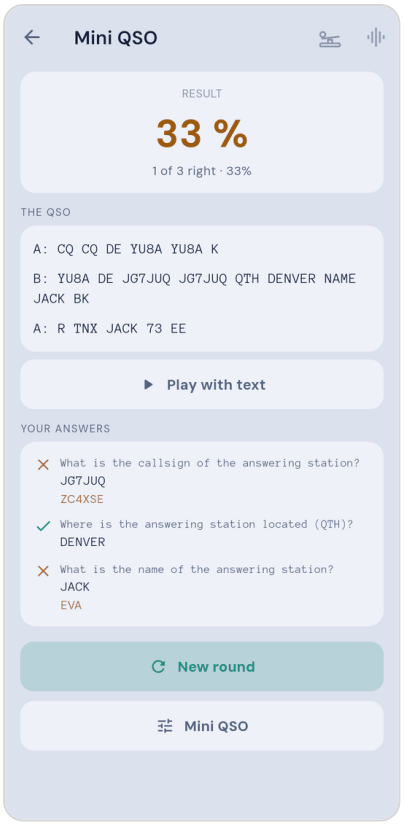
Result. The hit rate of the round, the whole exchange as “A: … / B: …” (with **Play with text**) and every question with the right answer and yours. The time counts for the daily goal.



Mini QSO: a question about the exchange

15.4 Morsel

A word puzzle like Wordle, but in CW.



Mini QSO: result with the whole exchange

- A game has **ten words**. One letter is revealed, and the word is played once in CW.
- Key the **whole word** back. A word pause submits it. `<err>` (eight dits) deletes the last character.
- Colours: **green** means right, **red** means wrong, and **grey** means the revealed letter was keyed wrong.
- After each miss the word is repeated **5 WPM slower**, down to 18 WPM. After 12 seconds without input it is played again. **Skip** gives up on a word.
- **Score:** your time + 5 seconds per guess. A skipped word costs 60 seconds. Lower is better.

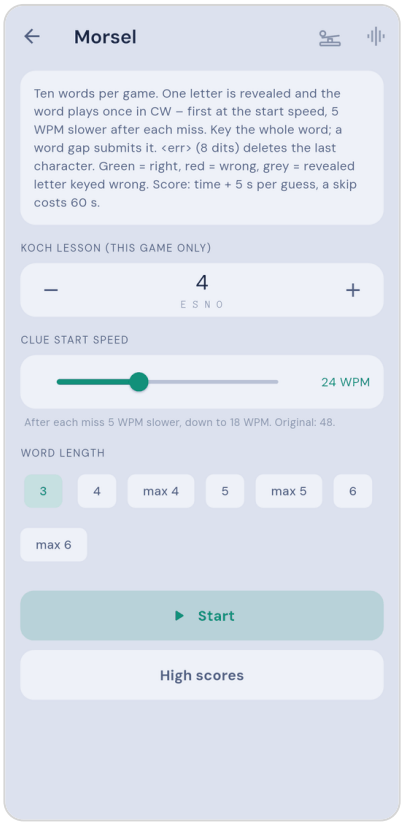
Setting	Meaning	Values
Koch lesson	For this game only. It starts at your Send lesson	2 to the end of the sequence
Word length	Exactly 3/4/5/6 letters, or at most 4/5/6	3, 4, max 4, 5, max 5, 6, max 6
Clue start speed	Speed of the first play	10–48 WPM (**48**, as on the Morserino)

Longer words need a higher Koch lesson. If too few words are available, the app tells you which lesson you need.

15.5 Morse Invaders

An arcade game. Characters from your Koch lesson fall down four lanes. **Key a character** to shoot the lowest invader showing it.

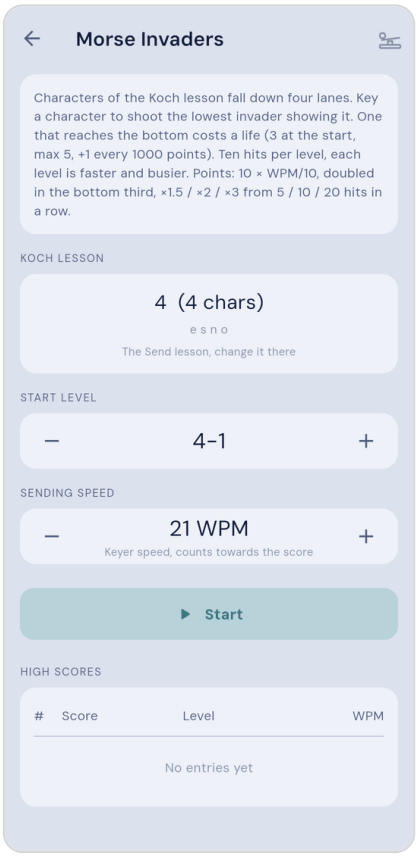
- A character that reaches the bottom costs a **life**. You start with 3 lives, can have at most 5, and gain one every 1000 points.
- After **ten hits** comes the next level. Each level is faster and busier.
- **Points:** 10 × WPM/10 per hit, doubled in the bottom third. Streaks add a multiplier: ×1.5 from 5, ×2 from 10 and ×3 from 20 hits in a row.
- **Pause** (the icon at the top right) stops the game. From there you can resume or go back to the menu. You can also change the speed with − and + during the game.



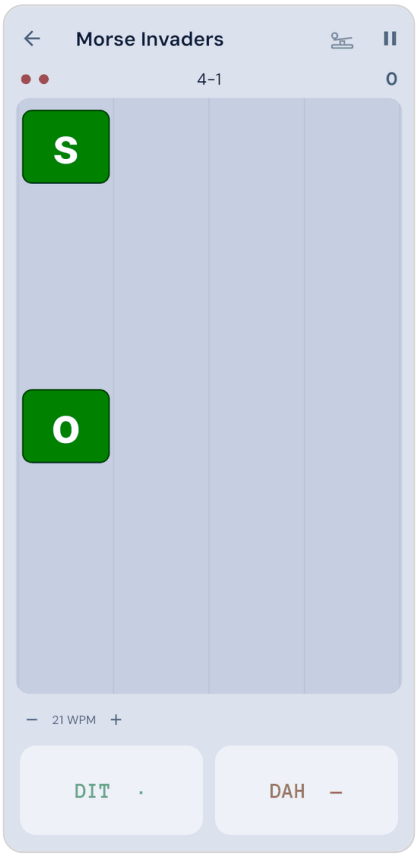
Morsel before the start

Setting	Meaning
Koch lesson	The lesson from **Send** (change it there)
Start level	The level you begin with
Sending speed	Keyer speed, counts towards the score

Results go into a high score list.



Morse Invaders before the start



Morse Invaders in play

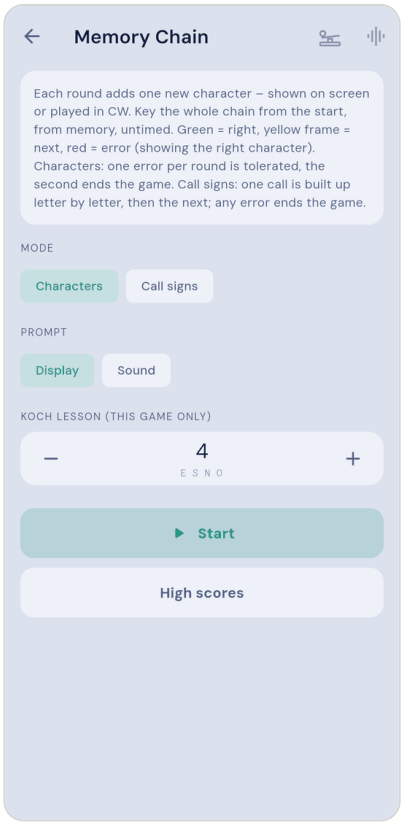
15.6 Memory Chain

A memory game. Each round adds **one character**, and you key the **whole chain** from the start, from memory. There is no time limit.

- **Characters mode:** random characters from your Koch lesson. One error per round is allowed, and the second ends the game.
- **Call signs mode:** a call sign is built up letter by letter, then the next one. Any error ends the game.
- **Prompt:** the new character is **shown** or **played** in CW (Sound).
- Colours: **green** means right, a **yellow frame** marks the next character, and **red** marks an error (with the right character).
- You set the **Koch lesson** for this game only before you start. It begins at your Send lesson. You can also change the speed with − and + during the game.

The high score list is kept per mode.

15.7 Trailblazer and Fox Hunt

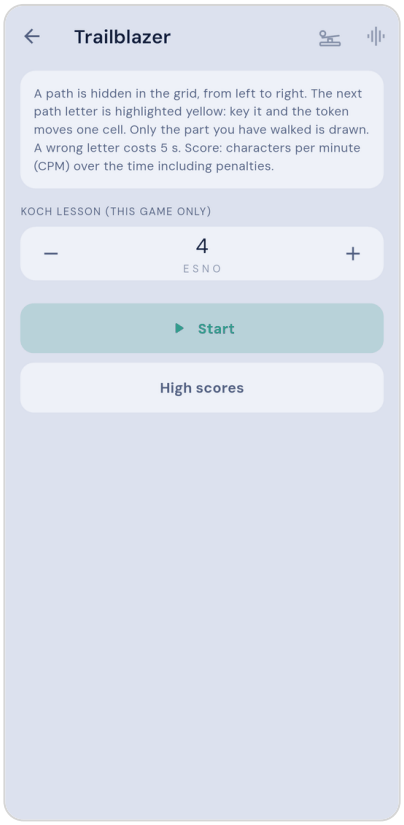


Memory Chain before the start

Two maze games on the same grid: **12 × 4 cells** filled with characters of your Koch lesson. A **path from left to right** is hidden in it. Only the part you have already walked is drawn, never the path ahead. The token (grey circle) moves one cell with every right answer; the game is solved at the right edge.

- **Trailblazer (sending):** The next path cell is **highlighted yellow**. Key its character.
- **Fox Hunt (listening):** You **hear** the character of the next path cell and key the **direction** to it, not the character itself. Example: you hear S, and the S cell is to the right of your token: you key **E**. The legend below the grid assigns a letter to each direction (↑ ↓ ← →): **N, S, W, E**, as far as your lesson has learned them. Otherwise the legend takes the first free letter of your lesson (**yellow frame**). The tone has the half-tone shift of the Send trainer. After 5.5 s without input it repeats; **Replay letter** plays it again at once.
- A wrong character (in Fox Hunt: the wrong direction, or a letter that is not in the legend) gives an error tone and costs **5 s**; a right one gives a confirmation tone.
- **Score:** characters per minute (**CPM**) = steps over the time including the penalty. Each game has its own high score list with seven places.
- You set the **Koch lesson** for this game only before you start. It begins at your Send lesson. You can also change the speed with − and + during the game. Fox Hunt plays the character at your keyer speed.

15.8 Fight the Pileup



Trailblazer before the start

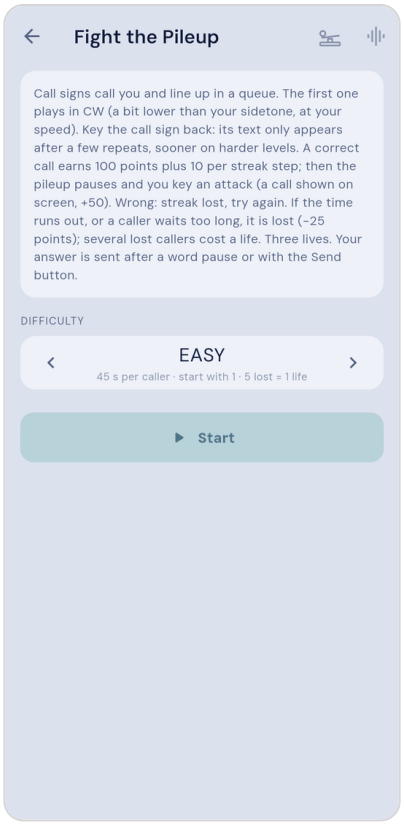
A pileup of stations calls you, and you have to copy them quickly. Call signs line up in a queue and the first one plays in CW, a little **lower** than your sidetone and at your keying speed. The call sign repeats until you answer. Key it back, and it is sent after a word pause, or at once with **Send**.

- **Reveal:** the text of the call sign appears only after a few repeats (3 on Easy and Normal, 2 on Hard, 1 on Expert). Until then you copy by ear.
- **Score:** a correct call gives 100 points plus 10 per streak step (x2, x3, and so on). A wrong answer ends the streak, and you try again.
- **Attack:** after each correct call the pileup **pauses**, and a call sign is shown on screen. Key it for 50 extra points; then the pileup goes on. In the firmware this attack goes to other players; the app has no multiplayer, so it only scores.
- **Lost callers:** if the time for the active caller runs out, you lose 25 points and the streak. A caller who waits too long in the queue leaves too. Several lost callers cost one life, and you have three lives.
- **Difficulty:** Easy, Normal, Hard and Expert change the time per caller (45 to 12 s), how fast new callers arrive, how many wait at the start and how many lost callers cost a life (5 to 2). Faster keying speeds up your own answers: you can change the speed with − and + during the game.

The call signs come from the same generator as in the other modes and follow the call sign settings from the ✲ sheet of Listen or Send (region, common prefixes only). There is no high score list, as in the firmware. When the game ends, you see score, defended and dropped callers, accuracy and best streak.

15.9 Radio Cave

A text adventure in an abandoned Cold War radio station: explore twelve rooms, find six items, repair the equipment and finish with a QSO with the remote station IR7. **Every command is keyed**; the game text is English.



Fight the Pileup before the start

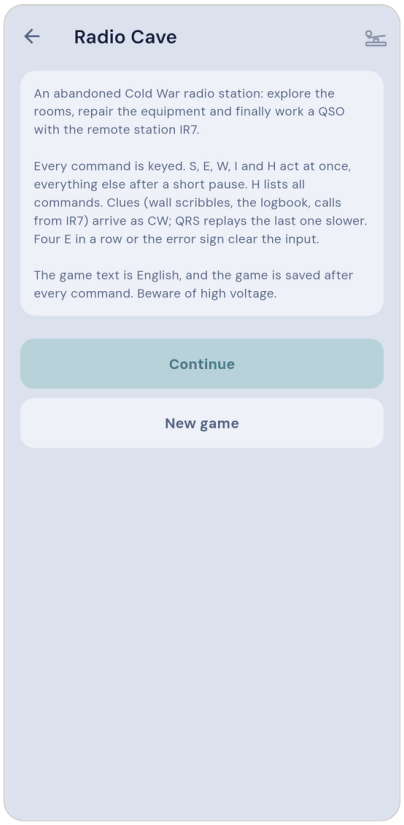
- The screen shows a small map of the cave (the current room is yellow), the room name, the text, the command line and below it the exits (N E S W), the items you carry (at most two) and the number of steps.
- **S, E, W, I** and **H** act at once. Everything else is sent after a short pause of silence (two and a half word spaces). **N** therefore waits, so that it can still become **NEW**.
- Commands: **N/S/E/W** go, **L** (+ object) look, **I** inventory, **T** (TAKE) and **D** (DROP) + item, **U** (USE) + item or thing, **U TX** / **U RX**, **F** (FIX) + item, **R** (READ) **MANUAL** or **LOG**, **K** (KEY) + word, **QRS**, **NEW**. **H** lists them. Items and things can be abbreviated (**T F** takes the fuel canister).
- **Clues come as CW:** the scribbles on the wall, the logbook, the manual and the calls from IR7. **QRS** replays the last clue at half speed. While a clue plays, keying is muted.
- Four **E** in a row or the error sign clear the input. After a prosign AS, KA, KN, VE or BK the game sees the letter S, A, N, E or B, as on the Morserino; SK counts as K.
- Handle **high voltage** with care: some actions are fatal. Then the game-over page offers a restart.
- The game is **saved after every command**. Next time you can continue or start a new game. Winning or dying deletes the save.
- Pitch, keyer mode and speed of your keying come from the settings; with − and
    - you change the speed during the game. The clues play at their own speed and at a fixed pitch of 696 Hz.

15.10 Text adventure

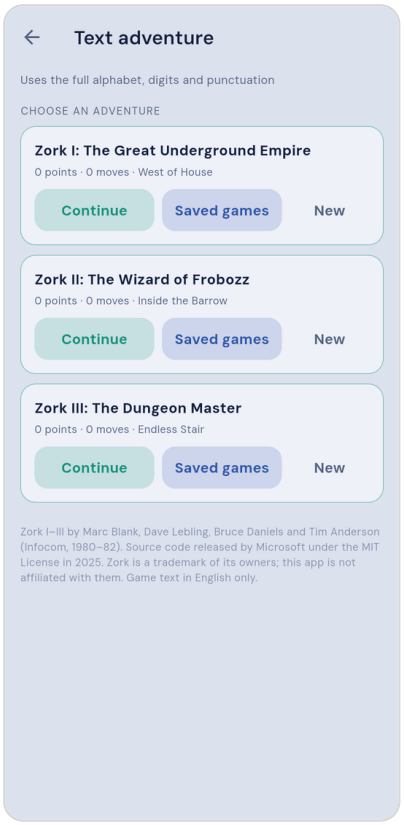
The three classic Infocom adventures **Zork I, II and III** (1980–82), played in CW: the game answers in Morse, and you enter your commands. Microsoft released the source code under the MIT License in 2025; the app plays the original story files with its own interpreter. The game text is in **English** only, and it uses the **full alphabet, digits and punctuation**, whatever your Koch lesson. Zork is a trademark of its owners; the app is not affiliated with them.

Selection. Each part shows its score, moves and last room. **Start** or **Continue** picks up where you left off. **Saved games** opens your saves. **New** starts over (after asking; your own saved games are kept).

The game screen. At the top the status line with room, score and moves, below it the transcript. The newest answer is played in CW, and the word that is sounding is highlighted. Below that:



Radio Cave before the start

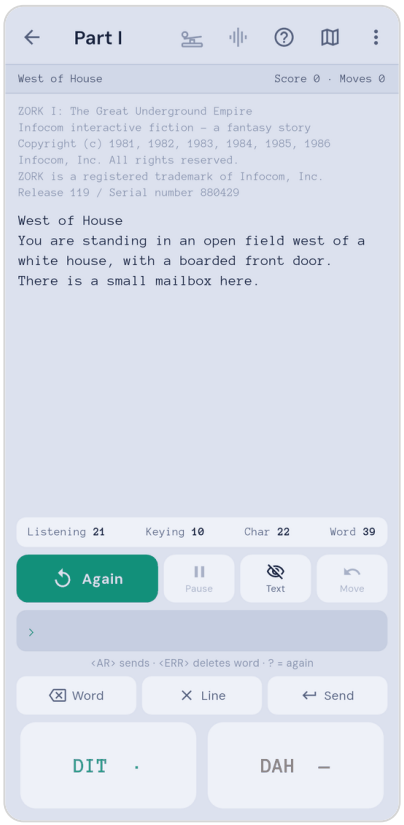


Choosing one of the three parts

Control	What it does
Speed bar	Listening and keying speed, character and word spacing; tap it for **Speed & spacing**
↻ **Again**	The sentence that is playing (in the gap right after a word: that word's sentence; after the end: the last one), then paused
Long press ↻ **Again**	The whole answer from the start (like sending `?`)
Tap a word	Only that word, then paused
Pause / **Resume**	Pauses playing; **Resume** goes on from the word where it paused – after ↻ **Again** from the next word
Text (eye)	Shows the whole answer at once; tap again to hide it, then the words reappear as they are played (e.g. with **Again**)
↶ **Move**	Takes back the last command (up to 20)

Keying commands. With the touch keyer at the bottom (keyer mode Straight: the key) or a connected Morse key (see Morse key). The decoder writes into the input line; the character being keyed is shown in orange as · and — after it. As soon as you start keying, the CW output stops.

- **End of a word:** a pause separates words. As in the sending learn mode, a word counts as finished after a pause of 2 × character spacing + 1 + word spacing / 8 dits at keying speed (straight key: word spacing + 1 dits after key-up).
- **Sending:** `<AR>` (·—·—·) sends the command at once. With the setting **" or K"**, a **K** as its own word also sends, i.e. after a word gap – none of the three games has a word K. The **Send** button always works.
- **Correcting:** `<ERR>` (8 dits; 7 or more count) deletes the last word, as does the ⌫ **Word** button. ✕ **Line** clears the whole input.
- A **?** on its own (then send) repeats the last answer and takes no move.
- Other prosigns are ignored; a character that isn't recognised shows as `*`.



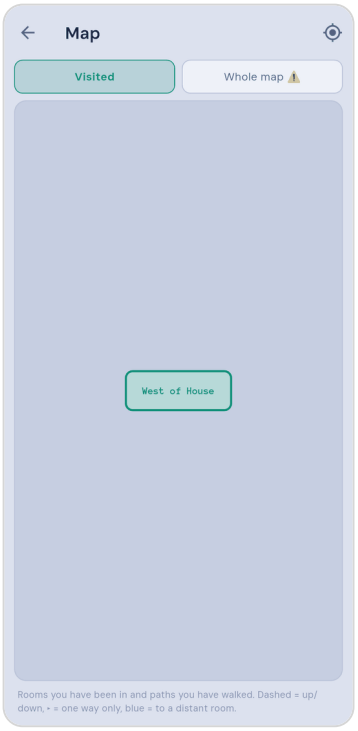
Game screen with touch keyer

With the setting **Input → Keyboard**, the on-screen keyboard replaces the dit/dah keys (for listening practice only): ␣ separates words, ⌫ deletes a character, ↵ sends. A connected Morse key works then too.

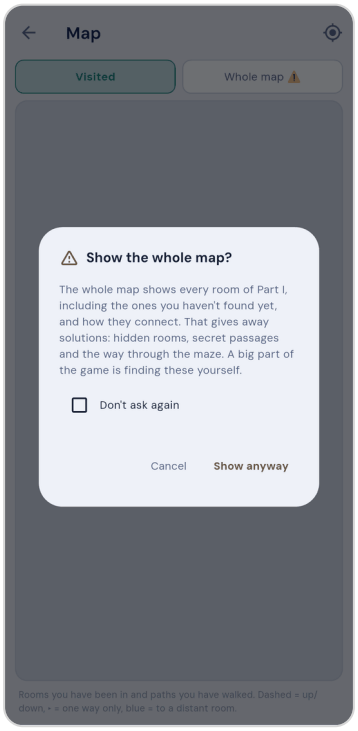
Map (?). The map icon in the title bar opens a map of the current part. It starts at the current room (orange outline); pinch to zoom, drag to move, ⌖ goes back to the current room. Rooms carry their

names from the game (English). Their positions are set by hand, because Zork isn't drawn to scale; the connections come from the story file.

- **Visited** (default): only rooms you have been in and paths you have walked – the map players used to draw on paper. No cheating. The paths are saved with the game and taken back by ↶ **Move**.
- **Whole map** ⚠: every room of the part, including the ones you haven't found yet (grey). That gives away solutions (hidden rooms, secret passages, the way through the maze), so the app asks every time. **Don't ask again** in the dialog turns the question off; turn it back on under ⋮ → Settings → **Map**.
- Dashed = up/down, ▸ = one way only. Distant connections (e.g. trap door, chimney) show as a blue note under the room ("→ Cellar") instead of a long line. Some paths only open during the game.
- In Part III the museum rooms appear three times side by side, labelled with the year (948 = the present, 776, 777): they are the same rooms at different times.



Map: Visited



Confirmation before the whole map



Map: Whole map

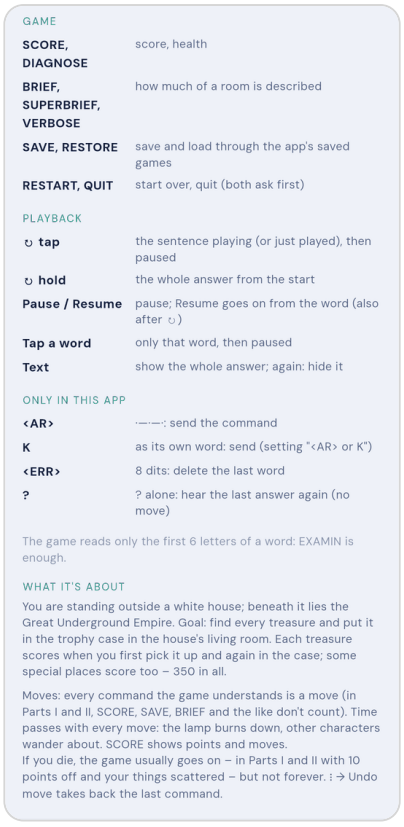
Command overview (?). The **?** in the title bar opens a list of the most important commands and, under **Playback**, the buttons above; at the bottom, **What it's about** briefly gives the story and goal of the current part, how points and moves are counted and what happens when you die. The short forms work in all three parts:

Command	Meaning
`N S E W, NE NW SE SW, U D, IN OUT`	Moving
`L`	LOOK: describe the room again (takes a move)

Command	Meaning
`I`	INVENTORY: what you carry
`Z`	WAIT: wait one move
`G`	AGAIN: repeat the last command
`OOPS word`	Replaces a word the game did not know
`TAKE`, `DROP`, `EXAMINE`, `READ`, `OPEN` …	Handling things (Zork has no `X` for EXAMINE)
`SCORE`, `SAVE`, `RESTORE`, `RESTART`, `QUIT`	Game commands
`<AR>`, `K`, `<ERR>`, `?`	Only in this app: send, send (as its own word, depending on the setting), delete the last word, again

The game reads only the first **6 letters** of a word (`EXAMIN` is enough). Separate several commands in one line with a full stop: `TAKE LAMP. N`.

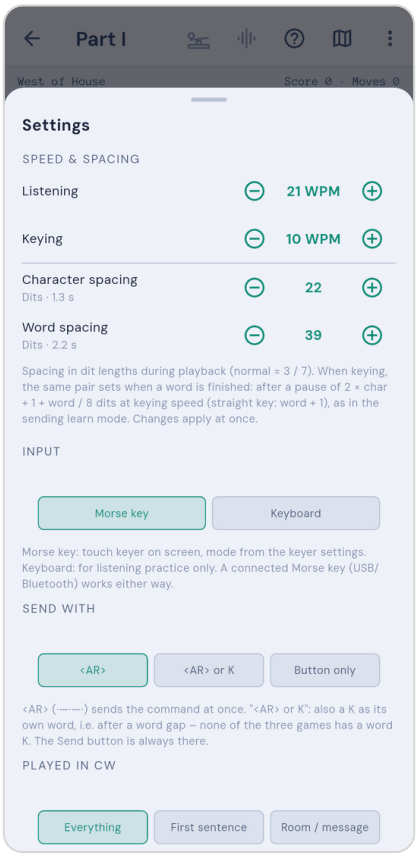
Settings (⋮ → Settings; shared by all three parts):



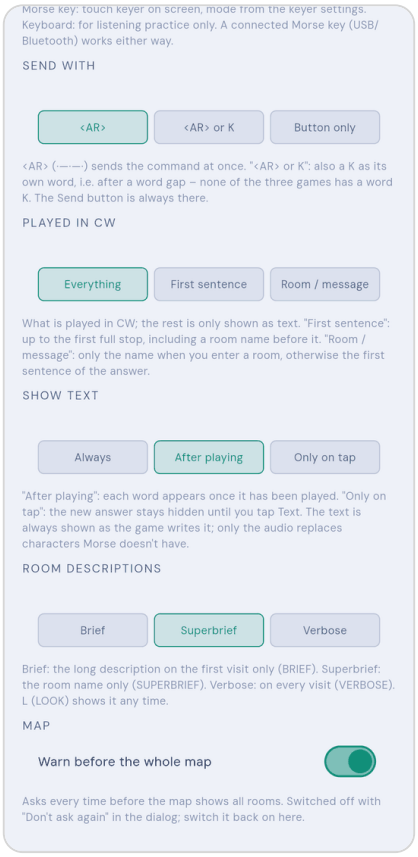
Command overview, “What it’s about” at the bottom

Setting	Meaning	Values
Listening	Speed of the CW output	10–60 WPM
Keying	Keyer speed for your input	**as listening**, 10–60 WPM
Character spacing	Pause between characters when playing; part of the word end when keying	3–45 dits
Word spacing	Pause between words; never below the character spacing; part of the word end when keying	6–105 dits
Input	Touch keyer or on-screen keyboard	**Morse key**, Keyboard
Send with	What sends a keyed command (the Send button always works)	**`<AR>`**, `<AR>` or K, Button only
Played in CW	What is played; the rest is only shown as text (italics)	**Everything**, First sentence, Room / message
Show text	When the new answer becomes readable	Always, **After playing**, Only on tap
Room descriptions	How much of a room is described	**Brief**, Superbrief, Verbose
Map: warn before the whole map	Asks before **Whole map** ⚠	**On**, Off

- The adventure takes speed and spacing from your **Listen** profile the first time you open it; after that they are its own values. You can also change them from the speed bar, and they apply at once, even while playing.
- **First sentence:** up to the first full stop, including a room name before it. **Room / message:** only the name when you enter a room, otherwise the first sentence of the answer.
- **After playing:** each word appears once it has been played. **Only on tap:** the new answer stays hidden until you tap **Text** or the box.
- **Brief** = BRIEF (the long description on the first visit only), **Superbrief** = SUPERBRIEF (the room name only), **Verbose** = VERBOSE.



Settings: speed & spacing, input



Settings: played in CW, show text, rooms, map

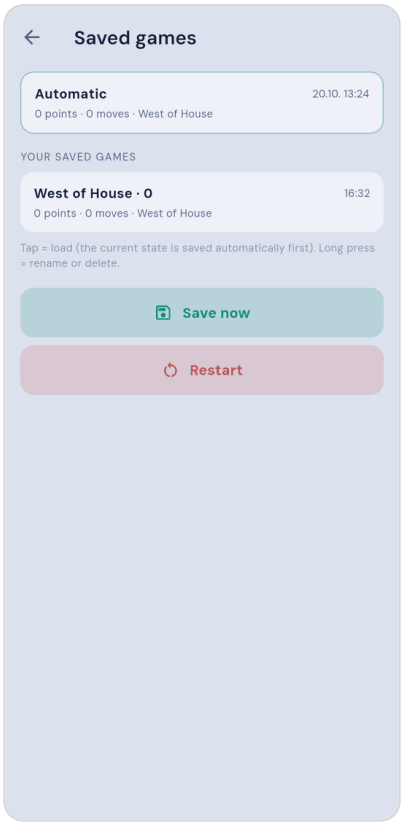
Punctuation in the audio. The screen always shows the text as the game writes it. Only for playing, characters that Morse doesn't have are replaced:

In the text	Played as
`'` `"` `(` `)` `[` `]` `*` `#` and others	left out
`!`	`.`
`;`	`,`
&	`AND`
Paragraph	double word gap

Saved games.

- **Automatic:** saved after every command and when you leave, one slot per part. Next time you continue exactly there.
- **Your own saves:** as many as you like per part, via ⋮ → **Save** or the game command `SAVE`. The name is suggested (room · score).
- **Loading:** via ⋮ → **Load**, **Saved games** in the selection, or the game command `RESTORE`. Tap to load; the current state is saved automatically first. Long press to rename or delete.

- **Restart:** ⋮ → **Restart** (the app asks first and offers to save before) or the game command `RESTART` (the game asks).
- **Undo (↶):** up to 20 commands while the game screen is open. The original doesn't have this; it mainly helps with typing mistakes.
- **End of the game:** after `QUIT` the app offers to load a saved game, take back the move, or restart.



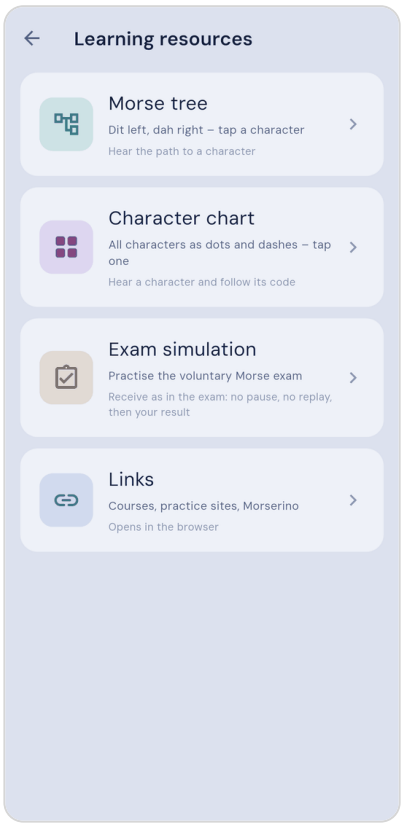
Saved games of one part

16 Learning resources

Under **Learn → Learning resources** you find help for learning CW that is not a training of its own.

16.1 Morse tree

The Morse tree shows all codes in one picture: from the dot at the top, a **dit goes left and a dah goes right**. The lines show it too: a dit is a dotted line, a dah a thick solid one. A character sits where its code ends, so you can read every code by following the path down to it. Letters are on the first four levels. **Digits and signs** adds the fifth level with digits and signs; the tree then scrolls sideways.

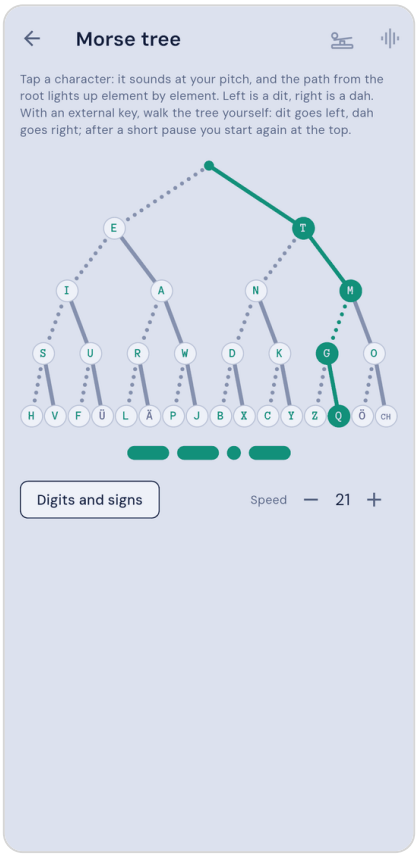


Learning resources: Morse tree, character chart, links

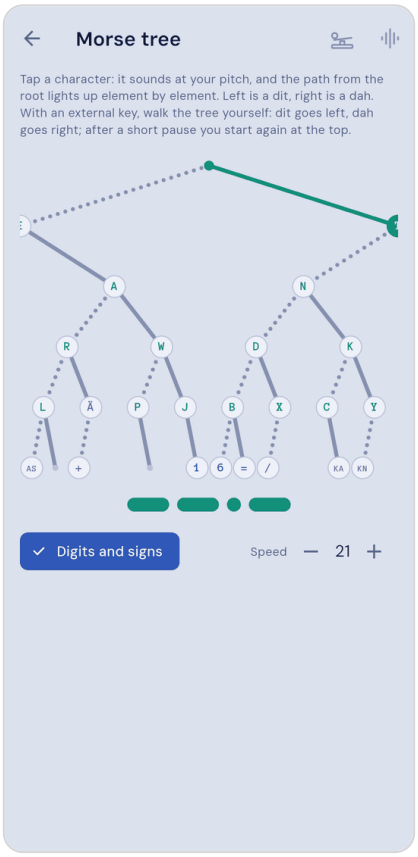
Tap a character to hear it. It sounds at your **pitch** (Settings → General) and at the speed shown below the tree. That speed starts at your **Listen** speed and can be changed here with **–** and **+** (10–60 WPM); the change applies only to this screen. While the character sounds, the path from the top lights up element by element: dit or dah is highlighted exactly when you hear it. Below the tree the code is shown once more as dits and dahs.

With an external key you walk the tree yourself: a dit moves one step to the left, a dah one step to the right, and the node reached lights up. If you pause for a moment, the node reached stays lit and you start again at the top. A step with no branch also starts over. The key must be learned in Settings → Keyer; the touch keys play no role here.

Note on CH (– – – –): the character at the bottom right (four dahs) is a historic national extra character, once common in German- and Scandinavian-speaking countries. It is obsolete in international radio today. Always send CH as two separate letters: C (– · – ·) and H (· · · ·). The app shows this note below the tree as soon as you tap the character. Ä, Ö and Ü can be played here too.

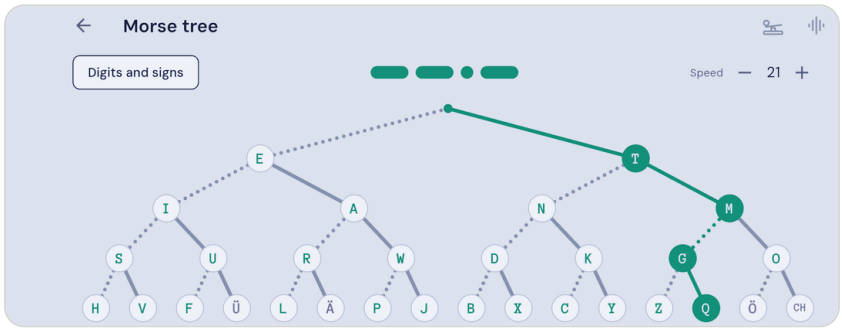


Morse tree: the path to Q lights up

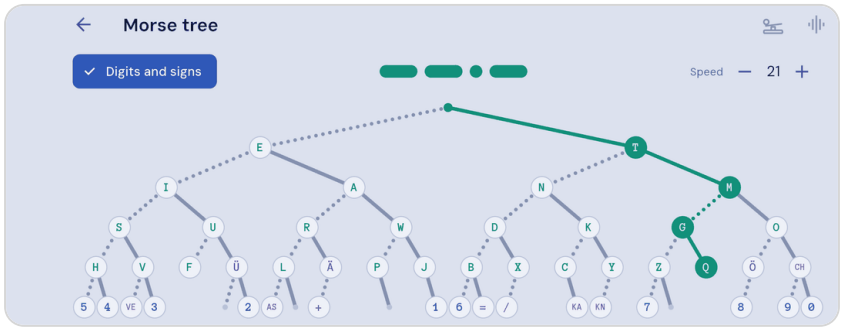


With digits and signs: the tree scrolls sideways

Turn the phone sideways for a bigger tree: the tree screen is the only one in the app that also works in landscape. There the controls sit in one row above the tree, and with **Digits and signs** the whole tree fits on the screen without scrolling. The rest of the app stays in portrait.



Landscape: letters



Landscape with digits and signs: the whole tree fits

The tree is drawn by the app from the same code table it uses for playing and decoding, so it always matches what you hear.

16.2 Character chart

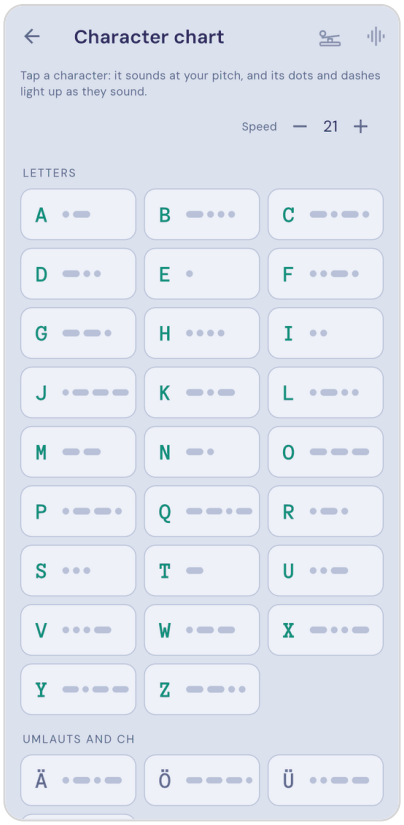
The character chart lists every character of the app with its code, grouped into **Letters**, **Umlauts and CH** (Ä, Ö, Ü, CH), **Digits**, **Punctuation** and **Prosigns** (SK, KN, KA, AS, VE, BK). Each code is drawn as dots and dashes.

Tap a character: it sounds at your pitch, and its dots and dashes light up as they sound. The speed starts at the value from the Listen training; **−** and **+** change it for this view only. Set the pitch under Settings → General. CH is a historic extra character; tapping it shows a note to always send it as the two letters C and H.

16.3 Exam simulation

Some countries offer a voluntary Morse exam. Under **Learn → Learning resources → Exam simulation** you can practise its receive part in the exam's own format and see whether you would pass. The app does not replace the official exam: the values come from public sources and the real exam may differ.

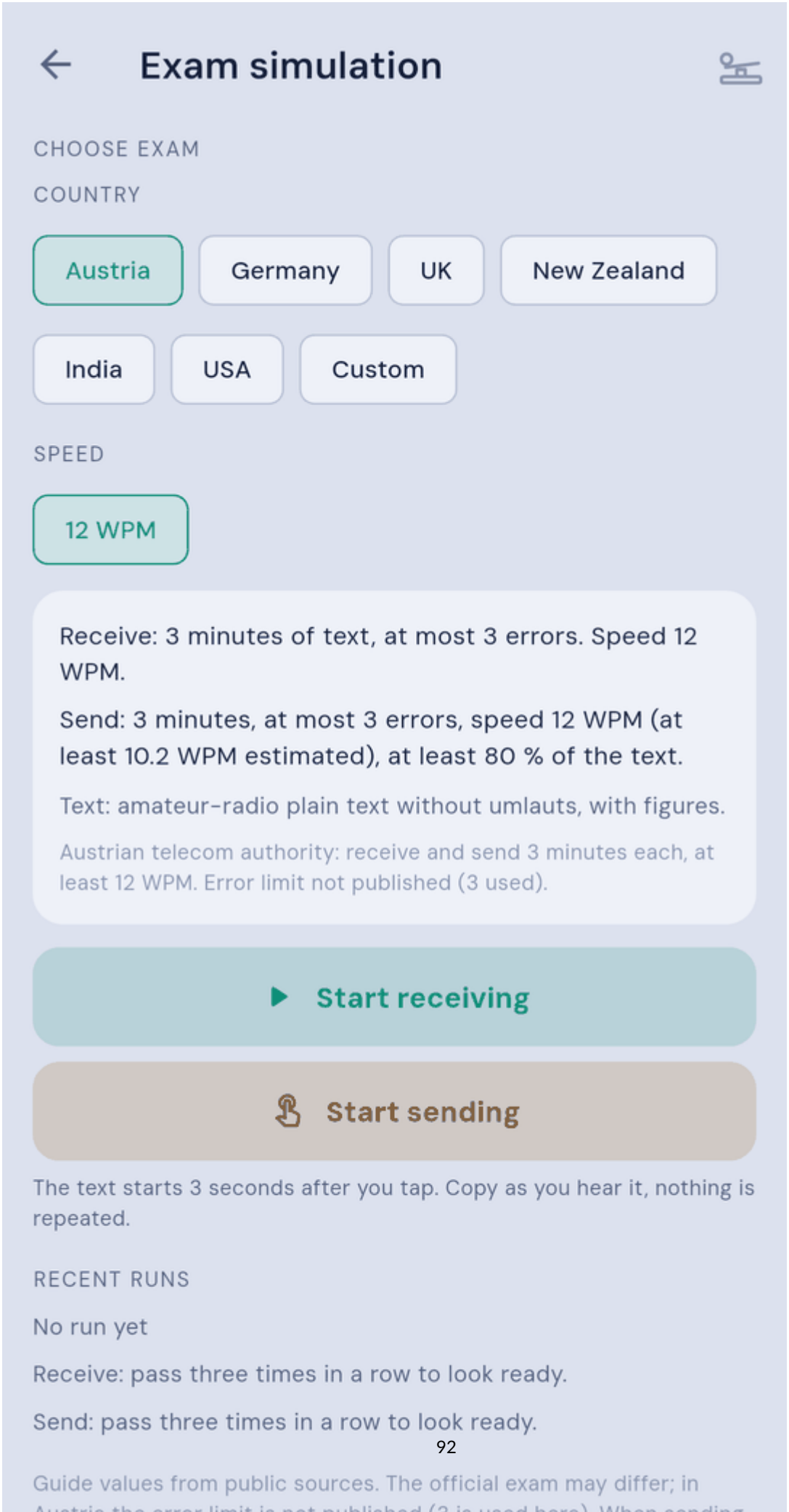
You first choose a **country**, then the **speed**:



Character chart: a character lights up while it plays

Choice	Variants and rules
Austria	12 WPM, 3 minutes, at most 3 errors (the official limit is not published)
Germany	5 WPM Farnsworth (characters at 9 WPM, long pauses), 5 WPM or 12 WPM, 3 minutes, at most 4 errors, with `. , ? = /` and AR (type `+`)
UK	5, 10, 12, 15, 20, 25 or 30 WPM; plain text 3 minutes (at most 4 errors) or **Figures**: groups of five figures for 1 minute (at most 3 errors)
New Zealand	5 WPM, 3 minutes, at most 4 errors, up to 5 attempts
India	5 or 8 WPM, 1 minute without a mistake
USA	ARRL Code Proficiency: 10 to 40 WPM, 1 minute without a mistake, receive only
Custom	your own profile, see below

::: {.shots one}


Exam simulation
CHOOSE EXAM
COUNTRY
Austria
Germany
UK
New Zealand
India
USA
Custom
SPEED
12 WPM
Receive: 3 minutes of text, at most 3 errors. Speed 12 WPM.
Send: 3 minutes, at most 3 errors, speed 12 WPM (at least 10.2 WPM estimated), at least 80 % of the text.
Text: amateur-radio plain text without umlauts, with figures.
Austrian telecom authority: receive and send 3 minutes each, at least 12 WPM. Error limit not published (3 used).
Start receiving
Start sending
The text starts 3 seconds after you tap. Copy as you hear it, nothing is repeated.
RECENT RUNS
No run yet
Receive: pass three times in a row to look ready.
Send: pass three times in a row to look ready.
Guide values from public sources. The official exam may differ; in

The card below the choice shows the rules of the chosen exam for both parts (receiving and sending), the source they come from and the kind of text. The plain text is amateur-radio text with figures: **German** (without umlauts) for Austria and Germany, **English** for the UK, New Zealand, India and the USA, with call signs and towns of the country in the generated QSO lines. In the custom profile you choose the language. The text is as long in Morse as the exam part lasts (a text of "WPM × 5 characters" would be shorter, because spaces and short characters go quickly). Figure and letter groups are groups of five characters.

16.3.1 Custom profile

Custom lets you define an exam yourself, for receiving and sending alike: **overall speed** (5 to 40 WPM), **character speed** (from the overall speed up to 40 WPM; a higher value gives Farnsworth pauses when receiving), **minutes** (1 to 10), the allowed **errors** (0 to 10), the **retries** of a failed part (0 to 5, see *Retries* above) and the **kind of text**: plain text (with switches for punctuation and AR at the end), figure groups or groups of letters and figures. The settings are saved. Your runs are listed per combination of settings, so "ready" refers to exactly this exam.

Start receiving begins after a 3 second countdown. The text plays once, without a pause and without a replay, while you copy it into the text field. When it ends you have 30 seconds left to correct your copy (or tap **Done**). Spaces and case do not matter. Every wrong, missing or extra character counts as one error.

The result page shows passed or not passed, the number of errors against the limit, the characters you copied wrong, the text that was sent (wrong characters in red) and your copy. **Recent runs** lists your last five runs per exam. After three passes in a row the app tells you that you look ready for the receive part.

The speed and spacing of the exam are set by the profile; your other settings (such as the pitch) are used.

16.3.2 Sending

Start sending on the exam page (or **On to sending** after a receive run) opens the send part. You first see the text and the rules. **Start sending** begins a 3 second countdown, then the time of the exam runs (three minutes) and you key the shown text with the touch paddle or a Morse key, in the keyer mode set under **Settings → Keyer**. With a paddle, a **WPM** slider sets the keyer speed: it starts at the exam speed (at 5 WPM Farnsworth at the character speed of 9 WPM) and you may send faster, up to 40 WPM, but not slower; the setting is remembered per exam. With a straight key there is no slider. The text you have keyed so far is highlighted, and below it you see what the app decoded. A correction (eight dots) deletes the last character, like a correction on paper. **Done** ends the run early.

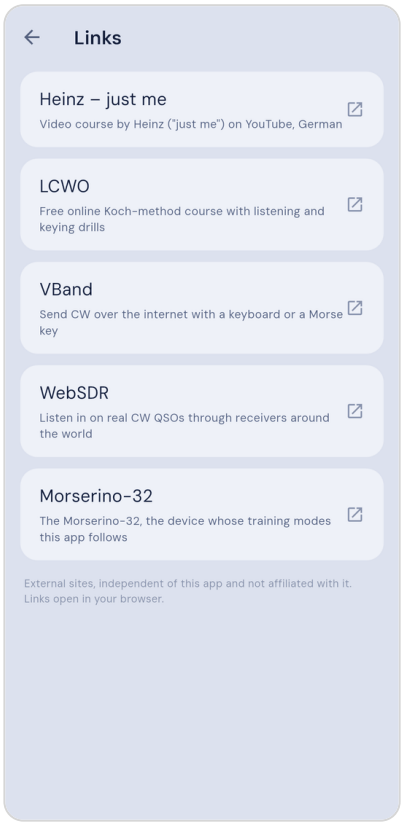
The result page shows three checks: the **errors** (substituted, missing or extra characters, against the limit), how much of the text you **reached** (at least 80 %) and your **speed**, estimated from the time between the first and the last element keyed (at least 85 % of the exam speed, measured in real Morse length, so figure groups count as they sound). All three must be fine to pass. Speed and amount of text are estimates by the app and not part of the official rules; an examiner also judges

rhythm and readability. Wrong characters are shown in red in the text, and the decoded text is listed below. The runs of both parts appear under **Recent runs**.

Retries. Where the exam allows a retry of a failed part (Germany: one per part, New Zealand: up to five attempts in all), the result page shows **Attempt 1 of 2** and the button changes to **Retry**. A new text is used for every attempt. When the retries are used up and the part is failed, it counts as failed, as in the exam, and the button becomes **New exam**. A passed part starts again with attempt 1. Receiving and sending count their attempts separately, and the counting starts again when you leave the screen.

16.4 Links

A list of external sites for learning CW: the video course by Heinz (“just me”) on YouTube, LCWO, VBand, WebSDR (listen in on real QSOs through online receivers) and the Morserino-32 homepage. A tap opens the site in your browser. These are independent offers that are not affiliated with this app; the app only links to them and shows nothing of their content.



Links to courses and practice sites

17 Morse key

17.1 Touch keyer

Every mode where you key shows two areas at the bottom: **DIT** (left) and **DAH** (right). In the **Straight** keyer mode there is a single **KEY** area that sounds for as long as you press it.

Hiding the touch keys. If you key with a real key on an adapter, you don't need the touch keys. To hide them, switch on **Settings → Keyer → Hide touch keys when an adapter is detected**. When you teach the dit and dah keys, the app also remembers your adapter and recognises it as soon as you plug it in. Without teaching on the adapter (or if you taught it earlier), it detects the adapter the first time you send a dit or dah with it. An ordinary USB or Bluetooth keyboard never counts. The touch keys then disappear in all modes; the keyer symbol in the title bar shows that an adapter is detected (see below). When you unplug the adapter, the touch keys come back. The "A real Morse key is better" tip no longer appears then.

The keyer symbol. The title bar of many screens (Listen, Send, practice, games, Morse chart and more) shows a symbol for the external key. It is dimmed while no adapter is detected and lit once one is. Tapping it opens the keyer settings. On screens where you do not key yourself you can also try the key out: dit and dah sound there with your Send settings (speed, pitch, keyer mode). It stays silent while a block runs. On the result page in Send the key chooses: **dah (–) = Next block**, **dit (·) = Finish**; the two buttons then carry a "-" or ".".

17.2 A real Morse key or straight key

A phone has no Morse key input. You need a small USB adapter that turns the key contacts into **key presses**:

- **vband** (hamradio.solutions/vband) is a widely used, ready-made USB adapter built exactly for this.
- **Homemade:** any small USB HID device that reports the dit and dah contacts as two different keys works. A ready-to-build example is diy-vband-adapter. It is a small adapter built from a microcontroller board with a 3.5 mm jack (several boards are supported) that sends the same keys as the vband adapter. It plugs straight into the phone with a USB-C cable.

Older phones with micro-USB need a USB OTG adapter.

Settings → USB CW adapter has a short hint with two links that open both of the above in your browser: the ready-made vband adapter and the instructions for the homemade one.

17.2.1 Learning the dit and dah keys

This tells the app which key your adapter sends for dit and which for dah:

1. Plug in the adapter.
2. Go to **Settings → USB CW adapter → Learn dit/dah keys**.
3. When “Press Dit key …” appears, press the **dit key**. When “Press Dah key …” appears, press the **dah key**.
4. “Saved” confirms it. The detected keys are listed under **Dit** and **Dah**; below them, under **Adapter**, the name of the device with its USB id (vendor:product, e.g. `Name (1a86:7523)`). The small key symbol in front is lit while the adapter is plugged in, and dimmed (faint text) when the app only knows it but it is not connected.

So there’s no list of supported adapters. The app learns whatever keys your adapter sends.

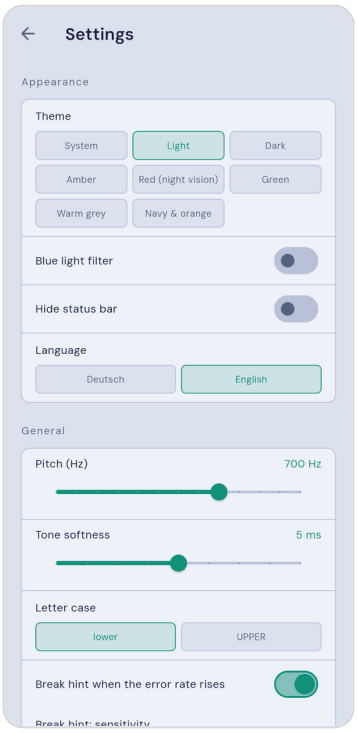
17.2.2 Analyze key events

If an adapter doesn’t behave as expected, use **Settings → Analyze Key Events**. Press **Start analyzer**, then press the keys. The app lists every key event it receives, so you can see whether the adapter sends anything and what it sends. **Stop analyzer** ends the display.

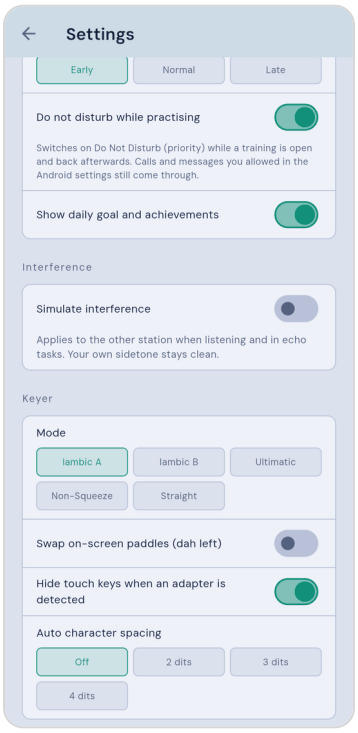
18 Settings

Open the global settings with the gear at the top right of the home screen. They apply to the whole app. Anything that concerns only one training is in that training's ⚙ sheet.

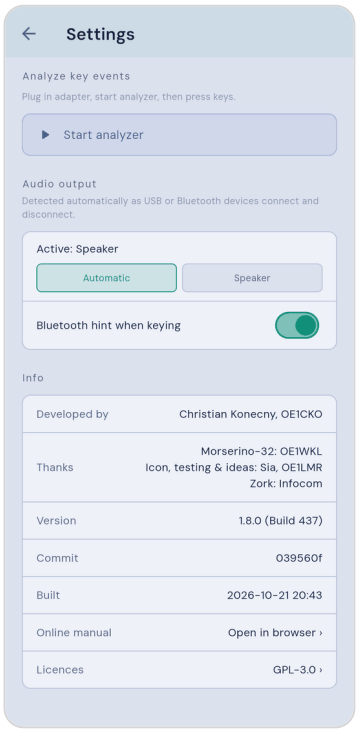
Changes apply immediately. Right after you change a switch, a choice or a slider (in the settings and in the ⚙ sheets), a bar at the bottom shows "Changed: …" with **Undo** for a few seconds. One tap on Undo restores the previous value, for example after touching a control by accident while scrolling. A newer change replaces the bar, so Undo always refers to the last change.



Settings: Appearance, General



Interference and Keyer



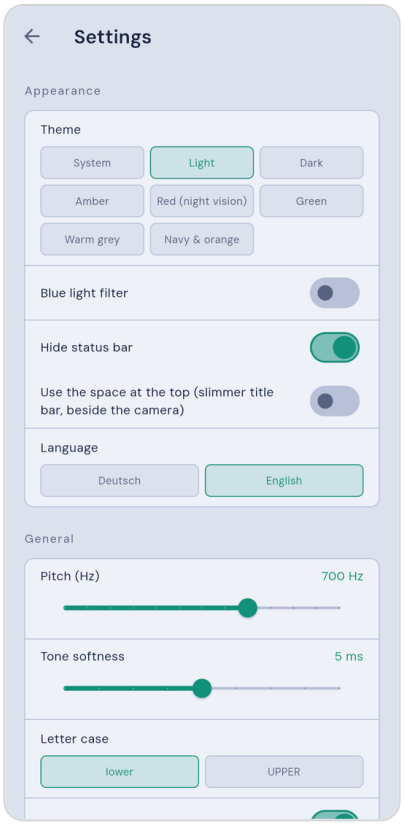
USB CW adapter, key events, audio output, Info

18.1 Appearance

Setting	Meaning	Values
Theme	Light or dark look, plus five more dark themes	**System** / Light / Dark / Amber / Red / Green / Warm grey / Navy & orange
Blue light filter	Warm colour filter over the whole app (light and dark), easier on the eyes in the evening	**Off** / On

Setting	Meaning	Values
Filter strength	How strongly blue is reduced (only shown while the filter is on)	10–100 % (**50 %**)
Hide status bar	Hides the Android status bar (clock, battery) so it does not clash with the blue light filter; swiping down from the top shows it briefly	**Off** / On
Use the space at the top	Only shown while the status bar is hidden. The title bar moves up into the freed strip beside the camera cut-out and gets slimmer (smaller text and icons, title independent of the system font size and shortened if needed so it does not end up behind the camera), so the content below gets more room. With a wide notch, a hole in a corner or too little room beside the camera (very narrow phones) the normal layout stays	**Off** / On
Language	App language. Until you choose one, it follows the phone's language: Deutsch on a German-language phone, English on any other	**Deutsch** / English (by phone)

18.2 General



Appearance: with “Hide status bar” on, the switch “Use the space at the top” appears

Setting	Meaning	Values
Pitch (Hz)	Frequency of the sidetone and of the played characters	300–900 Hz in 50 Hz steps (**600 Hz**)
Tone softness	Rise and fall time of the tone. Larger values sound softer and click less, especially on short dits	1–9 ms (**5 ms**)
Letter case	Show characters in lower or UPPER case. Display only	**lower** / UPPER
Break hint when the error rate rises	Suggests a break in Listen and Send when errors pile up or the hit rate drops clearly under the same conditions (see Break hint)	**On** / Off
Break hint: sensitivity	How early the hint comes (only shown while the hint is on)	Early / **Normal** / Late
Do not disturb while practising	While a training screen is open, switches Android's Do Not Disturb to "priority" and back afterwards, so other apps stay quiet. Calls and messages you allowed in the Android settings still come through. Needs Do Not Disturb access for APEX CW: the setting then shows a button to grant it, and one that opens the Android Do Not Disturb settings so you can switch it on yourself	On / **Off**
Show daily goal and achievements	Shows or hides the card on the home screen, and switches the recording of practice time on or off. Stored data is kept	**On** / Off

18.3 Interference

With **Settings → Interference** the other station sounds as if it came over a real band: with noise, static crackle, fading, a neighbouring station, drifting pitch and a "bad fist". That trains you to read a signal under poor conditions.

The interference only affects what the **other station** plays: the text when listening, the word when sending, Own texts, QSO Bot, WiFi Trx, Text Adventure, Morsel, Memory Chain, Trailblazer and Fox Hunt. Practicing a single character, the Morse chart and the Morse tree also play with interference while the switch is on, and show the wave icon as well. Your own sidetone while keying always stays clean. The statistics do not record whether you practiced with or without interference.

Quick switch: screens with another station show a wave icon in the app bar (next to statistics and the gear). Grey means off, coloured means on. A **short tap** switches the interference on or off, a **long press** opens these settings.

Setting	Meaning	Values
Simulate interference	Main switch	**Off** / On
Constant noise	Noise and QRM stay on for as long as an exercise, block or game is running, also between characters and under your own keying (your sidetone stays clean). When listening and sending this means a whole block, until the result appears; in Morsel, Memory Chain, Trailblazer, Fox Hunt and Fight the Pileup a game; in QSO Bot and WiFi Trx the session or connection; in Own texts only during playback. More realistic, but more tiring. Off: the interference is only heard while the other station is sending	**Off** / On
Preset	Ready-made mixes. You can change every slider of a preset directly; the change is kept for that preset. A small “Default” button (circular arrow) appears as soon as a preset differs from its defaults and resets it. “Custom” keeps its own values and has no reset	Custom / Light / HF evening / Pile-up
Noise (SNR)	Signal-to-noise ratio, measured in a fixed 2.4 kHz bandwidth. The noise swells and ebbs slowly and contains occasional crackle	0 % (no noise) to 100 % (SNR −10 dB), shown in dB
Noise colour	How bright the noise sounds	0 % dull (treble cut from about 800 Hz) to 100 % bright (about 3.2 kHz)
Receiver filter	Narrow CW filter around your pitch. The narrower it is, the more noise and neighbouring station it removes. At a very narrow setting the remaining noise sounds like a wavering tone	0 % wide to 100 % very narrow (Q 1.5 to 20)
QRM (other station)	A second station sends random Morse in runs of 3 to 12 characters (12 to 28 WPM), 100 to 350 Hz above or below your tone, then pauses. 100 % is as loud as your signal	0–100 %
QSB (fading)	The level varies slowly (periods of about 30 to 125 s) by up to about 8 dB	0–100 %
Pitch (drift)	The pitch wanders slowly, up to about ±30 Hz at 100 %. With a narrow filter the tone can drift out of the passband	0–100 %
Timing (bad fist)	Dits, dahs and gaps are uneven, the station sometimes hesitates	0–100 %

The presets start with (noise / QRM / QSB / pitch / timing / receiver filter, noise colour always 50 %):

Preset	Values
Light	15 / 0 / 10 / 0 / 0 / 30 %
HF evening	35 / 20 / 45 / 10 / 15 / 50 %
Pile-up	55 / 60 / 35 / 15 / 30 / 60 %

Try it plays four random groups of five with the speed and spacing of your Listen training, so you get an impression of how it sounds in the training. A known text like “CQ CQ DE …” would be much easier to read through noise than random groups.

18.4 Keyer

These settings apply wherever you key.

Setting	Meaning	Values
Mode	How the keyer reads the keys (see below)	**Iambic A** / Iambic B / Ultimatic / Non-Squeeze / Straight
Swap on-screen paddles	Puts **dah** on the left and **dit** on the right of the two touch buttons. Affects only the on-screen buttons, not the keys of an adapter (those stay as you taught them, see Learning the dit and dah keys). Not shown for Straight	**Off** / On
Hide touch keys when an adapter is detected	Hides the touch keys in all modes once an adapter has sent a dit or dah; the keyer symbol in the title bar shows that an adapter is detected (see Touch keyer)	**Off** / On
CurtisB dit timing	Iambic B only: from what percentage of a dit a press on the other key is already stored	0–100 % in steps of 5 (**75 %**)
CurtisB dah timing	The same for dahs	0–100 % in steps of 5 (**45 %**)
Auto character spacing	Enforces a minimum pause between characters so they don’t run together. Not available for Straight	**Off** / 2 / 3 / 4 dits

Setting	Meaning	Values
Straight key start speed	Straight only (replaces Auto character spacing): first estimate for the speed measurement, see Straight key	5–40 WPM (**15**)

The keyer modes

- **Iambic A**: if you hold both keys (“squeeze”), dits and dahs alternate. When you let go, the keyer stops after the current element.
- **Iambic B**: like A, but the keyer remembers a press on the other key that comes during an element, and adds that element (Curtis B behaviour). The CurtisB settings control from when this applies. 0 % means during the whole element, and 100 % means practically like Iambic A.
- **Ultimatic**: when both keys are pressed, the **last** one pressed wins and repeats for as long as it is held.
- **Non-Squeeze**: for single-lever keys or those switching over. Squeezing both keys produces no alternating sequence.
- **Straight**: a straight key. The tone is on for as long as the key is pressed. With touch, a single **KEY** area appears. With an adapter, the dit contact acts as the key. The speed is measured, see below.

18.4.1 Straight key: automatic speed

With the straight key you don’t set a speed. The app measures it from your keying, like the Morserino: from the length of your dits and dahs (a running average) it derives the speed, the dit/dah threshold and the character and word gaps. If you key more slowly, longer pauses are allowed automatically.

- The first estimate is the **Straight key start speed** (Settings → Keyer). If you key much more slowly, set it to your speed, otherwise a slow S can be read as three E at first.
- After about two to four characters the measurement has settled. The very first character can still be misread.
- The measurement restarts every time you enter a screen.
- The **WPM slider** is disabled and moves by itself to your measured speed (Keyer, WiFi Trx, Echo Trainer, QSO Bot and others). It is not saved and does not change the keyer speed.
- WiFi Trx sends the measured speed along. The Echo Trainer statistics store it as your sending speed.

18.5 USB CW adapter and Analyze key events

See Learning the dit and dah keys and Analyze key events.

18.6 Audio output

Setting	Meaning	Values
Active	Shows where the sound is going right now	–
Output	**Automatic** follows whatever is currently plugged in or connected. The other options fix the output. Only outputs that are currently available are offered	**Automatic** / Speaker / Wired/ USB / Bluetooth
Bluetooth hint when keying	Shows a hint above the keys when the sound plays over Bluetooth (delay about 100–250 ms)	on (default) / off

18.7 Info: version and build

Row	Meaning
Developed by	Christian Konecny, OE1CKO
Thanks	Morserino-32 (OE1WKL), app icon, testing and ideas (Sia, OE1LMR), Zork (Infocom)
Version	Version number and build number, for example “1.0.0 (Build 42)”
Commit	The exact source code state the app was built from
Built	Date and time of the build
Online manual	Tap to open this manual online in your browser (German or English, following the app language)
Licences	Tap to open the licence texts (see below)

When you report a problem, please include the version, commit and build time.

Licences: The app is free software under the GNU General Public License v3.0 (or later). It takes algorithms and data (word lists, abbreviations, call sign prefixes, QSO texts) from the Morserino-32 firmware by Willi Kraml, OE1WKL, which is also under the GPL-3.0. The source code is on GitHub. The licence page also shows the MIT License of Zork I–III, the SIL Open Font License of the fonts Anonymous Pro and DM Sans, the CC-BY-SA-4.0 licence of the word lists (derived from the FrequencyWords project), and the licences of the Flutter packages the app uses. Zork is a trademark of its owners; the app is not affiliated with them, nor with Infocom, Activision or Microsoft.

18.8 Koch sequence

You set the Koch sequence in the ⚙ sheet of **Listen** or **Send** when the **Koch lesson** character set is selected there. It applies to **all** trainings and games, though.

Sequence	Description
M32	The Morserino-32 order (51 steps): `m k r s u a p t l o w i . n j e f 0 y v , g 5 / q 9 z h 3 8 b ? 4 2 7 c 1 d 6 x – = SK + AS KN KA VE BK @ :`
LCWO	The lcwo.net order
CW Academy	The CW Academy (CWops) order
LICW	The Long Island CW Club order with an **entry point** (see below)
Custom	Your own order

LICW entry point (0–13): in the LICW course, students join a “carousel” at different points. The entry point rotates the sequence so that it starts there.

Custom: enter the characters in the order you want to learn them. Duplicates are ignored, and the app shows how many characters it found. The default is `esno0tqr5ucd9al8ix1myj7h4gvkfz3b.6/w2p?`, the order of the YouTube Morse course by “Heinz – just me” (playlist).

Prosigns: the M32, LCWO, CW Academy and LICW sequences contain the six prosigns SK, AS, KN, KA, VE and BK exactly where the firmware puts them (mostly at the end). A prosign is a lesson step of its own: it appears as a two-letter tile, is played as one character (not as two letters) and turns up in random groups. Words and abbreviations contain no prosigns. They are left out of the games (Morsel, Memory Chain, Maze, Invaders), where every character has to fit into one box. The LICW carousel follows the firmware rule: its window holds the prosigns that fall inside its 18 or 44 characters (SK and BK); the others do not occur there. In **Custom** and in the **practice set** you write a prosign in angle brackets, e.g. `esn<KA>t<SK>` (allowed: AS, KA, KN, SK, VE, BK); it counts as one character. Without brackets `KA` is two letters.

18.9 Morserino terms

The app gives the Morserino-32 menu items plain names. If you come from the Morserino or read its manual, this table helps:

Morserino menu	In the app
Interchar Spc	Character spacing
InterWord Spc	Word spacing
Random Groups	Group characters
Length Rnd Gr	Group length
Length Words	Max word length
Length Abbrev	Max abbreviation length
Length Calls	Max call sign length

Morserino menu	In the app
Calls Region	Region (under Call signs)
Max # of Words	Groups per block / Words per block
Stop<Next>Rep	Stop after each group / Stop after each word
Koch Sequence	Koch sequence
Practice Set	Practice set
Boost Practice	Boost practice set
Echo Prompt	Prompt
Echo Repeats	Repeats
Echo Speed Max	Sending speed (max)
Tone Shift	Tone shift
Confrm. Tone	Confirmation tone
AutoChar Spc	Auto character spacing
Output Case	Letter case
Keyer Mode	Mode (under Keyer)
CurtisB DitT% / DahT%	CurtisB dit timing / dah timing

The names of the keyer modes (Iambic A, Ultimatic …) and of the Koch sequences (M32, LCWO …) are the same as on the Morserino.

19 What the app does not do (yet)

Some Morserino-32 features are missing from the app, because a phone doesn't have the hardware or Android handles it already. These are the rotary knob and buttons, the display, LoRa, ESP-NOW (and with it the multiplayer parts of the games), iCW/Ext Trx and keying a real transmitter, firmware updates and the WiFi setup page. Instead of the firmware's Practice Stats, the app has its own, more detailed character statistics.

Not done yet, but planned: the games Trailblazer, Fox Hunt, Radio Cave and Fight the Pileup.

20 Troubleshooting

The app doesn't key when I press my Morse key. Learn the dit and dah keys (see Learning the dit and dah keys). If nothing arrives, check with **Analyze key events** whether the adapter sends anything at all.

Sound comes out of the wrong device. Fix the output under **Settings → Audio output**.

When sending, I hear my tone late. This is almost always Bluetooth headphones. Use wired headphones or the speaker.

The next Koch character doesn't come. Open the  statistics in Listen. The characters without a tick are holding up the unlock. Usually it's a character with too few attempts or recent errors. See When the next Koch character comes.

The pauses never get shorter. That's intended while you work through the Koch sequence (see Pauses and speed). You can shorten them yourself at any time with the **speed and spacing** card.

The decoder only writes garbage. Check pitch, bandwidth and threshold (see Decoder settings). Only turn on the monitor tone with headphones.